

Visual Network Tool: Individual Belief Networks from LLM-Guided Interviews and a Visual Canvas

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Abstract

Belief network theories make predictions on individual and collective levels, but are typically measured only at the collective level. We introduce a new tool to measure individual-level belief networks that combines semi-structured interviews and a visual canvas. This Visual Network Tool addresses several limitations of existing measurement approaches and enables fast and scalable measurement of the differences in content and structure of individual belief networks. Participants complete a semi-structured interview about their beliefs on a given issue guided by an LLM, evaluate LLM-extracted statements that summarize the beliefs they have expressed, and construct their individual belief networks by assigning *supporting* and *conflicting* relations between these statements on a visual canvas. We validate the tool in a two-wave study on beliefs related to meat-eating. Belief network content and structure show satisfactory reliability across the two measurement points, and we find support for face, concurrent, and criterion validity. We show that interconnected beliefs are more likely to persist over time, consistent with predictions from belief network theories. This tool opens up new empirical possibilities to test theories of belief dynamics.

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1 Introduction

A growing number of theories of belief dynamics build on the concept of individual and collective belief networks (Bizyaeva et al., 2022; Brandt & Sleegers, 2021; Dalege et al., 2016, 2025; Goldberg & Stein, 2018; Rodriguez et al., 2016; Schweighofer et al., 2020; Vlasceanu et al., 2024). In these accounts, a belief about an issue is embedded within a system of other beliefs about related issues, motivations, experiences, and behaviors. This representation makes it possible to apply ideas and methods from network science to formulate and test theoretical hypotheses about individual and collective beliefs. For example, this representation allows studying why some people’s belief networks are more resilient (Scheffer et al., 2022), why some beliefs are more difficult to change than others (Brandt & Vallabha, 2023; Turner-Zwinkels & Brandt, 2022), how belief networks change to adapt to internal and external pressures (Dalege & van der Does, 2022; Dalege et al., 2016; Turner-Zwinkels & Brandt, 2022), and how network structure relates to decision and action (Brandt et al., 2019; Dalege et al., 2017; Hasan et al., 2025). In this paper, we use the term “belief” to refer broadly to the different types of cognitive content individuals associate with an issue, including those described above.

Despite all this progress and theorizing about belief networks, a major limitation has been the difficulty of measuring the beliefs of an individual. Most existing studies rely on belief networks inferred from population-level data, where relationships between beliefs are obtained from correlations between individuals (Boutyline & Vaisey, 2017; Dalege et al., 2025; Turner-Zwinkels et al., 2021). While valuable for understanding how beliefs affect each other and predict behavior (Chambon et al., 2023; Dalege et al., 2017) the population-level approach has important conceptual limitations. Population-level estimates of belief networks might not represent individual belief networks well (Brandt, 2022; Brandt & Morgan, 2022), obscuring the variation in network content and structure at the individual level. For example, some individuals might have more while others have fewer beliefs about a given issue. Take the topic of meat eating. This might prompt some people to think mostly about the taste of meat (Hopwood

et al., 2021; Piazza et al., 2015), while others might relate the topic to a wider set of beliefs, for example, about the health effects of meat-eating and animal welfare concerns (Hopwood et al., 2020). In addition, individual belief networks can be more or less interconnected. For example, people with more political knowledge might have more interconnected beliefs about political issues than those with less knowledge (Boutyline & Vaisey, 2017; Brandt, 2022). Individual differences in the number of beliefs, their interconnectedness, and other aspects of network structure could constrain the way people adjust their beliefs in light of new evidence and changes of related beliefs, although current evidence is mixed (Brandt & Vallabha, 2023).

There are two main ways in which researchers have tried to measure individual belief networks. The first method—longitudinal measurement—asks an individual about the same beliefs repeatedly over several time points, and infers relationships between these beliefs from the patterns of their co-change over time (Chambon et al., 2022, 2023). However, estimating individual-level belief networks from longitudinal data relies on individual-level belief change. This is a challenge because beliefs tend to be relatively stable over time (Kiley & Vaisey, 2020). In addition, estimating temporal networks requires many observations (Mansueto et al., 2023). This is not ideal since individual belief systems might change because the individual is forced to respond to the same questions repeatedly (Bartels, 1999). The second method to measure individual belief networks—pairwise measurement—asks an individual to make conceptual similarity judgments about pairs of beliefs and treats these similarity values as the network (Brandt, 2022). However, this method can become cognitively and financially costly when researchers are interested in the relationships between more than a few beliefs, since participants are asked about each pair of beliefs separately. In addition, the mere act of asking about a connection between two beliefs might induce a relationship between them even if a participant did not think about it before (Tourangeau et al., 2000). Finally, pairwise comparisons by definition allow only for evaluating each relationship in isolation from the whole belief system.

More generally, most previous methods to obtain belief networks relied on pre-specified questionnaires. For example, beliefs about genetically modified food (GM) might be measured using a questionnaire designed to reflect the theoretically assumed moral and social beliefs relevant to GM food (Dalege & van der Does, 2022; Dalege et al., 2025). While enabling easy

inter-individual comparisons, such pre-specified questionnaires assume or impose the same number of beliefs on all participants, even when some participants might have never considered a particular belief before, or when some have many more or more nuanced beliefs than specified in the questionnaire. Furthermore, the mere act of asking about a belief can change a person's belief system, as individuals may report a belief they did not have before being asked (Schwarz, 1999; Tourangeau et al., 2000). Therefore, belief networks measured with pre-specified questionnaires can misrepresent individual belief systems and mask genuine individual differences. As a result, they might be less predictive of belief change and behavior than they could be.

Here we introduce a new measurement methodology, the Visual Network (VN) Tool, to elicit individual belief networks through a combination of LLM-guided semi-structured interviews and a visual canvas on which participants indicate how their beliefs relate to each other. The VN Tool mitigates some of the conceptual and practical shortcomings of previous measurement approaches. Rather than relying on pre-specified questionnaire items, the VN Tool elicits beliefs in an open-ended and non-suggestive manner, allowing participants to report only the beliefs they consider relevant, in their own words. In addition, rather than asking about the relationship between each belief pair and possibly imposing connections that were not considered before, the VN Tool allows participants to indicate the relationships that they consider relevant on a visual canvas without a specific prompt to focus on any two beliefs. As a result, the VN Tool is less likely than existing methods to affect participants' belief systems through the process of measurement and more likely to produce data that is useful for describing and predicting belief dynamics. Lastly, it reduces the cognitive and financial costs of measurement, opening doors to large-scale empirical studies of individual belief networks and further theory development and testing. In the next sections, we describe the tool and investigate its' reliability and validity to measure individual belief networks.

2 Description of the Visual Network Tool

The VN Tool consists of three main components (Figure 1): an LLM-guided semi-structured interview, LLM-extraction of belief statements that participants rate by their accuracy and relevance, and a visual canvas on which participants arrange and connect the belief statements to create their personal belief networks. In addition, we develop a training procedure that can be used to teach participants, if needed, how to draw connections on the canvas. For validation purposes in this study, we also include sections in which participants evaluate pairwise relationships for a subset of belief pairs, assess how accurately various network representations describe their beliefs, and answer several more standardized questions about their beliefs. While the tool can be used to explore beliefs in any domain, here we apply it to the topic of meat-eating. We chose this topic because beliefs and behaviors related to meat-eating are likely to be relatively stable over time (Beardsworth & Bryman, 2004), enabling us to assess the test-retest reliability of the tool. The interface design for the VN Tool is documented in Section S1.1.



Figure 1: Overview of the three main components of the Visual Network (VN) Tool. **A.** Semi-structured LLM-guided interview (here on beliefs about meat-eating). **B.** Participants rate LLM-extracted belief statements regarding their *accuracy* and *relevance*. **C.** Participants create *supporting* (purple) and *conflicting* (orange) connections between belief statements on a visual canvas. Video material documenting the VN Tool, code to deploy the VN Tool, and supporting scripts can be found at <https://github.com/victor-m-p/beliefs-narratives-networks>.¹

2.1 LLM Semi-structured Interview

The first part of the survey instrument consists of a semi-structured LLM-guided interview. The LLM is instructed to never be suggestive or ask about specific beliefs (e.g., about animal welfare). Instead, it is asked to explore participants' beliefs about issues, motivations, experiences, and behaviors related to meat-eating, with the objective to cover at least beliefs related to (1) the meat-eating habits of the participant, (2) the personal motivations that the participant has to eat meat or avoid eating meat, (3) the meat-eating habits of the social contacts of the participant, and (4) the motivations social contacts have to eat meat or avoid eating meat.

In the default setting, after being asked a written question by the LLM, a participant speaks their answer. This answer is recorded and then transcribed with Whisper (Radford et al., 2023). The interview transcript is then updated with this new Whisper-transcribed answer. The LLM is sent the updated transcript, and then generates the next interview question. This is repeated until the LLM asks a total of eight questions. A participant can choose to type their answers instead of speaking, in which case the transcription step is skipped and the interview process otherwise remains the same. The interview protocol and design is inspired by recent work in conducting semi-structured interviews with LLMs (OpenResearch, 2025b; Park et al., 2024; Wuttke et al., 2025), and participants rate their experience with this part of the VN Tool very positively (see Figure S10A). The choice of LLM is discussed in S1.10, and the LLM-prompt for the interview is shown in Section S1.10.1.

2.2 Extraction and Rating of Belief Statements

The second phase of the VN Tool starts with an LLM-extraction of belief statements about participants' meat-eating beliefs, extracted from a transcript of the interview. LLMs have been successfully used in previous research to extract deeply held attitudes and supporting beliefs (Velez et al., 2025). Here we provide the complete transcript of questions and answers from the interview as input to the LLM and instruct it to return a maximum of N belief statements (here set to 10). The LLM-prompt for extracting belief statements is documented in Section S1.10.2.

¹ All screenshots show the answers of the lead author, completing the survey on 2026-02-04.

Participants evaluate their personally extracted belief statements on two dimensions, “accuracy” and “relevance”, both measured on a slider-bar scale from 0 (“Not at all”) to 100 (“Very”; see Section S1.1.2). Belief statements rated by participants as at least “Accurate” (60 on the 101-point scale) are mapped on a visual canvas, described next. We treat this threshold as a design choice to retain participant-endorsed statements while excluding clearly mismatched ones. Participant ratings of relevance are used only to validate that the LLM can separate out beliefs that are specifically relevant to the participant. Additionally, relevance might be investigated further in terms of whether it correlates with network properties such as centrality, but this is not reported on here. Overall, participants rated LLM-extracted belief statements as both accurate and relevant (see Figure S10B).

2.3 Visual Canvas Task

Using a visual canvas, participants annotate network relations between their personally extracted belief statements that they have accepted as accurate. Such visual maps have been shown to be an effective and versatile method to obtain similarity judgments (Goldstone, 1994; Hout et al., 2013). Most typically used in the study of concepts (Richie et al., 2020), the method was also applied in the political domain (Lammers et al., 2017). In our setup, participants first arrange their belief statements such that beliefs that “go together” are closer to each other, and beliefs that do “not go together” are further apart on the canvas. They do so by dragging their belief statements into the square canvas from the side of the screen (see Figure S3). This serves two purposes. First, participants can organize their beliefs and consider how they relate before coding relationships. Second, the spatial organization may be analyzed as a proxy for how closely beliefs are related.

Participants then assign connections between belief statements that “support each other” (Figure S4) and that “conflict with each other” (Figure S5). We chose “support” and “conflict” to capture feelings of consistency and inconsistency between beliefs while introducing as few assumptions as possible about the nature of the relationship (e.g., logical, semantic, causal, or social). To create a connection, participants click on one of their belief statements, which highlights it, and then they click on another one of their belief statements, which completes the

connection. To remove a connection, participants can follow the same approach.

3 Methods

3.1 Participants

We applied the VN Tool in a two-wave study on the same participants. The two waves were separated by a relatively short one-week interval to avoid confounds due to actual changes in the belief systems of participants (DeVellis & Thorpe, 2021). Participants were recruited from the crowd-sourcing platform Prolific. The first wave (started on December 9, 2025) was completed by 247 participants, of which 217 participants completed the second wave (started on December 16, 2025). After excluding participants who failed the most obvious attention check (asking participants whether it is accurate that their friends often go out to eat seahorse) or had incomplete records, we were left with 210 participants included in the analyses reported next.

3.2 Procedure

In both waves of the study, participants completed the three core components of the VN Tool described above in Sections 2.1, 2.2, and 2.3. In wave 1, participants additionally completed three training examples before the visual canvas task (see Section 3.2.1). Participants also completed a short questionnaire about their meat-eating frequency and (potential) feelings of dissonance (see Section 3.2.4). In wave 2, participants were exposed to one training example, randomly sampled from the set of three training examples administered in wave 1. After the three main parts of the VN Tool, participants were asked about the relationship between a subset of belief pairs in a pairwise task (see Section 3.2.2). Then they compared and rated the accuracy of various network representations of their meat-eating beliefs (see Section 3.2.3). Participants completed the same questionnaire as in the first wave along with two adjusted questionnaires on meat-eating (see Section 3.2.4). Below we describe the canvas training task (Section 3.2.1, the pairwise evaluation task (Section 3.2.2), the network comparison task (Section 3.2.3), and the administered questionnaires (Section 3.2.4).

3.2.1 Canvas Training. Before participants completed the visual canvas task (see Section 2.3) they received a canvas training task. This task allows participants to become familiar with creating “supporting” and “conflicting” connections between beliefs. Participants first read a short vignette describing a person with some goal (e.g., learning a language) along with motivations and challenges. Then they (1) spatially arranged the provided belief statements, (2) annotated supporting connections, and (3) annotated conflicting connections. As such, the canvas training task exactly mirrors the downstream visual canvas task but it uses vignettes with pre-defined belief statements instead of the personally extracted meat-eating beliefs of participants. We designed the vignettes and the belief statements such that some statement pairs should be coded as “supporting” and some should be coded as “conflicting”. This allows us to provide live feedback to participants during training, and to estimate the degree to which participants create connections that comply with this notion of consistency (see Methods Section 3.4.2 and Results Section 4.7.2).

3.2.2 Pairwise Evaluation. We implemented a pairwise evaluation task to elicit belief relations in an alternative way to the proposed visual canvas task. For each participant, we sampled up to nine pairs of belief statements (*nodes*) from the network they created in the visual canvas task (see Section 2.3). When possible, we included three pairs that the participant coded as “supporting”, three “conflicting” pairs, and three unconnected pairs.

For each of these belief pairs, participants went through the following two steps. First, an LLM-interviewer asked the participant an open question about the two statements (e.g., “In your own words, how does [x] and [y] relate to each other, if at all?”; not analyzed in the present paper). After the participant responded, the LLM-interviewer asked a follow-up question (e.g., “Thank you for explaining that. If you had to choose, how would you describe the way these two things relate overall?”). For this second question, participants were given three options indicating support, conflict, or no connection. These pairwise belief relations can be directly compared to the canvas belief relations. This allows us to compare how consistently participants report belief relations using alternative methods (see Methods Section 3.3.7 and Results Section 4.5).

3.2.3 Network Comparison. To evaluate whether participants visually recognize their canvas-created meat-eating belief network as an accurate representation, we designed a network comparison task. Participants evaluated and compared four network representations of their meat-eating beliefs. The first was the network they created using the visual canvas task. The second network representation we extracted during the survey with an LLM (see Section 3.3.1) based on participant interview transcripts and belief statements. The third and fourth network representations were perturbed versions of the first two, created by randomly flipping connections between belief statements (e.g., from “supporting” to “conflicting”). Participants were shown two of these four networks at a time. They rated how accurately each network representation described their meat-eating beliefs. Because there were four network representations, and participants rated all pairs, this resulted in six pairs and three ratings for each network representation per participant. The task is designed to establish face validity of the network representation (see Methods Section 3.3.6 and Results Section 4.4).

3.2.4 Questionnaires. In both waves of the study, participants answered questionnaires about their meat-eating frequency (past, present, future), and about feelings of personal and social dissonance in the meat-eating domain. In the second wave, participants completed two standardized questionnaires about meat-eating motivations. The Vegetarian Motives Inventory (Hopwood et al., 2020) measures health, environment, and animal welfare motivations to eat less meat. The Motives to Eat Meat Inventory (Hopwood et al., 2021) categorizes common meat-eating motivations into four categories (natural, necessary, normal, and nice). We document these in Section S1.2, but they are not analyzed in the present paper.

3.3 Analysis

3.3.1 LLM Network Extraction. In addition to the belief networks that participants created on the visual canvas, we extracted individual belief networks using an LLM. We used the gpt-4.1-2025-04-14 model for all LLM-tasks described in this paper, but this can easily be changed to other proprietary or open-source model options. We chose this model to achieve the best possible understanding while ensuring low response times during the study

(see Section S1.10). The LLM-extraction of belief connections happened during the study in the background, after participants had completed the LLM interview and rated their LLM-extracted belief statements. The LLM received the belief statements that participants rated as accurate and the entire interview transcript as input. The LLM returned a set of connections between belief statements that are plausibly supporting or conflicting for the participant given the provided context (prompt in Section S1.10.3). We use the LLM-extracted networks in three ways. Participants rate the LLM-extracted networks alongside their own canvas-created networks in the analysis of *face validity* (see Section 3.3.6). The LLM-extracted connections are compared directly to the canvas-created connections in the analysis of *concurrent validity* (see Section 3.3.7). Lastly, the networks based on LLM-extracted connections are used as alternatives to the canvas-created networks in the analysis of *criterion validity* (see Section 3.3.8).

3.3.2 Mapping Statements to Topics. To compare the semantic content within and between participants, a mapping between belief statements and some comparable set of topics is needed. This paragraph describes how we arrived at one *selected* topic model that we primarily report and interpret, and a set of *candidate* topic models that we include for robustness.

We first selected four candidate sentence embedding models for BERTopic modeling (Groendorst, 2022). The all-MiniLM-L6-v2 and bert-base-nli-mean-tokens (Reimers & Gurevych, 2019) models have been widely used in NLP research, while bge-large-en-v1.5 (Xiao et al., 2024) and qwen3-embedding-0.6b (Zhang et al., 2025) are more modern sentence embedding models. All models were run through the Sentence Transformers (Reimers & Gurevych, 2019) framework from Hugging Face (Wolf et al., 2020).

We ran a grid search over hyperparameters to identify well-fitting topic model configurations. The varied parameters were: `n_neighbors` $\in \{15, 30, 45\}$ and `n_components` $\in \{2, 10\}$ for UMAP (McInnes et al., 2018), and `min_cluster_size` $\in \{25, 50, 75\}$ for HDBSCAN (McInnes et al., 2017). All other parameters were fixed to default values. Prior to dimensionality reduction we L2-normalized all sentence embeddings and removed English stop-words. Given the combinations of three hyper-parameters and four sentence embedding models this resulted in $3 \times 3 \times 2 \times 4 = 72$ topic model fits in total.

To identify the set of *candidate* topic models, we first removed all fits in which the com-

bined fraction of statements assigned to the outlier topic (topic -1) and the first topic (topic 0) exceeded 40% of all statements. Next, we removed all fits that had less than 5 topics and more than 40 topics based on prior expectations about the dimensionality of the meat-eating domain. Among the 40 remaining fits, we selected the 10 that had the highest DBCV metric score (Moulavi et al., 2014). We use these 10 *candidate* fits throughout the paper to assess the robustness of analyses that rely on a mapping between belief statements and topics.

To identify one *selected* bertopic fit to primarily report and interpret, we manually inspected each of the 10 *candidate* fits based on whether their topics reasonably separated four expected topics: “family eats meat”, “protein (I want, need)”, “animal welfare (concerned)”, and “environment (concerned)”. We selected the all-MiniLM-L6-v2 model run with `n_neighbors = 45`, `n_components = 10`, and `min_cluster_size = 75`. An overview of the 16 topics for this *selected* fit is shown in Section S1.6.

Across all analyses that involve *topics* (see Sections 3.3.5, 3.3.8, 3.4.1), we represent topic assignments as binary presence or absence. BERTopic assigns each statement a probability distribution over topics, but we simply use the topic with the highest probability as the single topic for that statement. A topic is considered present for a participant if at least one of their belief statements is primarily assigned to that topic. Unless stated otherwise, all analyses include the outlier topic that BERTopic produces.

3.3.3 Belief Network Description. To provide an overview of the types of belief networks that participants created with the VN Tool we show a diverse set of belief networks and report on the number of belief nodes, belief edges, and types of belief edges that we observe. We investigate the relationship between the spatial distance of belief statements on the canvas and the types of relationships that participants annotated between these belief statements (see Results Section 4.1).

3.3.4 Reliability of Network Structure. Participants should produce similar networks in wave 1 and in wave 2. To investigate reliability of the network structure obtained with the VN Tool, we compute correlations between the number of belief statements that were extracted from each participant interview, and the number of connections that each participant annotated

on the canvas, within individuals across the two waves of the study (see Results Section 4.2).

3.3.5 Reliability of Semantic Content and Connections. Participants should produce semantically similar content over short time intervals. The shared set of *topics* (see Section 3.3.2) allow us to compare semantic similarity both within participants and between participants across the two waves of the study. We quantify the semantic similarity between topic vectors with the phi coefficient which measures the correlation between two binary variables. The within-participant phi coefficient (ϕ_{within}) is calculated between the topic vectors of the same participant for wave 1 and wave 2. The between-participant phi coefficient (ϕ_{between}) is calculated between each participant’s topic vector in wave 1 and every other participant’s topic vector in wave 2. The between-participant phi coefficient is a baseline which we expect to be lower than the within-participant phi coefficient.

We apply the same approach to the *edges* of the topic networks (see Section 3.3.8; but unweighted). Each participant’s network is represented as a binary vector covering all possible connections among the topics. Connections are undirected and unweighted, and include self-loops. The within-participant phi coefficient for topic edges (ϕ_{within}) is calculated between the topic connection sets for the same participant across wave 1 and wave 2, and the between-participant phi coefficient for topic edges (ϕ_{between}) is calculated between each participant’s wave-1 vector and every other participant’s wave-2 vector, again serving as a baseline (see Results Section 4.3).

3.3.6 Face Validity of Network Representation. Participants should rate the networks they create themselves with the VN Tool as more accurate than alternative representations. To assess the face validity of the belief networks that we obtain with the VN Tool, we use the ratings from the network comparison task that we administered in the second wave of the study (see Section 3.2.3). In this task, participants rated four network representations on how accurately they represent their meat-eating beliefs (101-point scale anchored at “Not at all”, “Moderately”, “Perfectly”). Each participant made three comparisons of each network representation because ratings were done in all possible pairs. We calculate one average rating per participant per network representation (see Results Section 4.4).

3.3.7 Concurrent Validity of Canvas Connections. Participants should annotate similar relationships between belief statements using the VN Tool and alternative measurement approaches. To assess concurrent validity, we compare whether the visual canvas task (described in Section 2.3) elicits similar network relationships (supporting, conflicting) between beliefs compared to two alternative methods. First, we compare the canvas connections with connections that participants indicated in a pairwise evaluation task (see Section 3.2.2). Second, we compare them to connections extracted with an LLM during the study (see Section 3.3.1). For both comparisons, we report a 3×3 confusion matrix between supporting connections, conflicting connections, and belief-pairs that are not connected (see Results Section 4.5). In the pairwise evaluation task, participants only annotated connections between a subset of belief pairs, and we limit the comparison between the canvas-created connections and the LLM-extracted connections to the same set to facilitate comparison.

3.3.8 Criterion Validity of Network Structure. A theoretical prediction is that topics that are more interconnected in the belief networks of individuals should be more stable over time (Dalege et al., 2017, 2025). We evaluate whether topics that are more interconnected in a participant’s wave 1 belief network are more likely to be present in their wave 2 belief network. First, we construct a topic-level network from the canvas-created network of each participant in wave 1. Each canvas connection between two statements maps to a connection between the topics those statements are assigned to. We define $w_i(T, T')$ as the number of statement-level connections between topics T and T' for participant i ; this includes within-topic connections ($T = T'$), which arise when two statements assigned to the same topic are connected on the canvas. The weighted degree of topic T in participant i ’s wave-1 network is

$$d_i(T) = w_i(T, T) + \sum_{T' \neq T} w_i(T, T')$$

This gives us one degree value per participant-topic pair (i, T) where T is a topic present in participant i ’s wave-1 network. In the analysis later, we binned the topic degrees into weakly ($d_i \in [0 - 1]$), moderately ($d_i \in [2 - 3]$), and highly interconnected topics ($d_i > 4$). We investigate whether more interconnected topics in wave 1 reappear more frequently in the participants’

belief statements in wave 2. We define,

$$y_i(T) = \begin{cases} 1 & \text{if topic } T \text{ is present in participant } i\text{'s wave-2 network,} \\ 0 & \text{otherwise.} \end{cases}$$

and compute $\bar{y}_{ib}$ for each participant i and degree bin b . This is the fraction of their wave-1 topics in bin b that are also present in wave 2. The *persistence rate* for bin b is then the mean of $\bar{y}_{ib}$ across the participants that have at least one wave-1 topic in that bin (see Results Section 4.6). This analysis does not distinguish between supporting and conflicting connections.

3.4 Additional Analysis

3.4.1 Collective Belief Network. We aggregate participant-level topic networks (see Sections 3.3.2 and 3.3.8) to create a collective belief network representation. We qualitatively analyze the topics and topic connections to assess whether our approach recovers meaningful and expected structure within the domain of meat-eating (see Results Section 4.7.1). We apply edge filtering to reveal meaningful connections (see Section S1.9).

3.4.2 Canvas Training. We designed canvas training examples (see Section 3.2.1) to assess whether training makes participants better at using the VN Tool to report supporting and conflicting relationships between their beliefs. Each belief statement for the training examples is either supporting or conflicting with the goal described in each vignette. For instance, one vignette introduces Alex who wants to learn a new language (see Section S1.1.4). The statement “Alex practices daily with an app” supports this goal. We define a simple consistency rule: statements that are both supporting or both conflicting with the goal described in the vignette should themselves be connected by a supporting edge, whereas statements with opposite signs should be connected by a conflicting edge. We calculate the participant-level average consistency with this rule across training rounds for both supporting and conflicting connections to assess whether training examples make participants more compliant with this logic. To test whether participants also applied this logic for the main canvas task concerning their own beliefs, we use an LLM to classify their belief statements into beliefs that encourage *more* or *less*

meat-eating (see Results Section 4.7.2).

4 Results

4.1 Belief Network Description

Participants created a diverse set of belief networks using the VN Tool. We show a sample of these in Figure 2. Each network consists of belief statements (network nodes), spatial positions on the canvas (x, y), and connections to other belief statements (“supporting”, “conflicting”). Some networks show clustering, often into categories of beliefs that encourage more meat eating or less meat eating (e.g., Figure 2A,B). We also observe belief networks that appear to be more organized around one central belief, with few connections between other beliefs (e.g., Figure 2C,D). Some networks are almost fully connected (e.g., Figure 2E), while some are very sparse (e.g., Figure 2F). In some cases, participants create both supporting and conflicting connections between the same two beliefs (e.g., the green connection in Figure 2F). Overall, this variation suggests substantial individual differences in how beliefs are organized, including in clustering, centralization, and density. On average, the belief networks consist of $M = 7.9$ ($SD = 1.56$, $\min = 3$, $\max = 10$) nodes with $M = 9.2$ ($SD = 4.51$, $\min = 2$, $\max = 29$) connections between them.

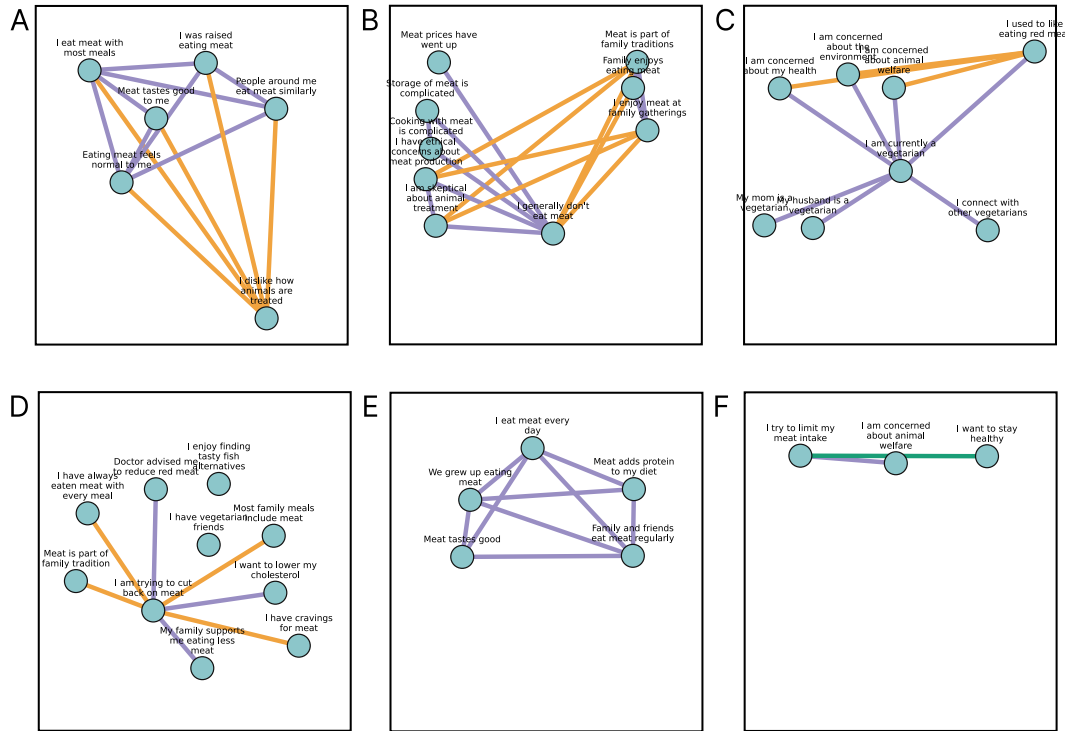


Figure 2: Belief networks on the topic of meat-eating elicited with the VN Tool, for six participants. Supporting connections are purple, conflicting connections are orange, and a node-pair with both connection types is shown in green. **A, B.** Networks with clusters. **C, D.** Networks with a highly central node. **E.** Nearly fully connected network. **F.** Sparse network. Each canvas appears like it does to participants during the study.

Participants create many more supporting than conflicting connections between their beliefs. 11% of the belief networks that we elicit have no “conflicting” connections (an example is Figure 2E) while 0% of networks have no supporting connections. Most networks have both supporting and conflicting connections (89%). The average number of supporting connections is $M = 5.8$ ($SD = 2.9$, $min = 1$, $max = 23$) and the average number of conflicting connections is $M = 3.4$ ($SD = 2.5$, $min = 0$, $max = 12$).

Closer proximity on the canvas is strongly associated with “supporting” connections (see e.g., Figure 2A, B). Across all participants and both waves, 31% of all possible belief-pairs are connected on the canvas on average. For the two nodes that a participant places closest to each other, we observe a “supporting” connection in 37% of the networks, while for the two nodes that a participant places furthest away from each other, we observe a “supporting connection” in only 12% of the networks. We observe a slight tendency in the opposite direction for “con-

flicting” connections, with 7% of the closest nodes classified by participants as “conflicting” and 14% of the most distant nodes classified as “conflicting” (see Figure S11). This also means that the overall rate at which a node-pair is connected on the canvas (either “supporting” or “conflicting”) drops from 45% for the pair that is closest to 28% for the pair that is furthest apart.

4.2 Reliability of Network Structure

The network structure that we extract with the VN Tool shows moderate reliability across the two waves of the study. The number of extracted belief nodes (excluding the limit of 10 maximum beliefs) is moderately reliable ($r = 0.38$). The number of network relationships that participants create shows a stronger correlation ($r = 0.54$). Figure 3 shows the reliability of network nodes and network edges. Table S4 additionally shows the correlation for interview lengths, the number of nodes extracted during the study (limited to 10), and split by each connection type (supporting, conflicting).

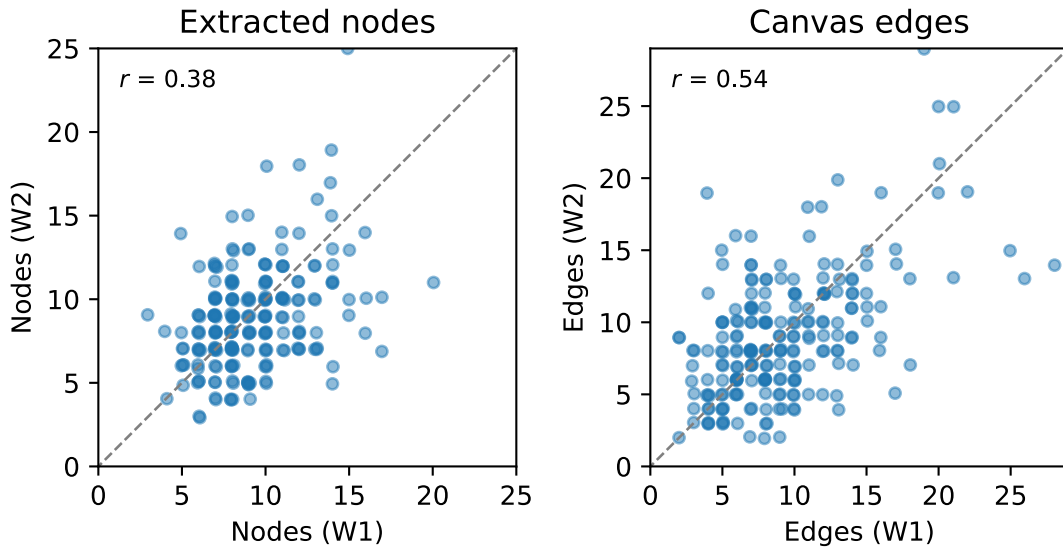


Figure 3: Test-retest reliability of extracted belief nodes and canvas edges. *Left*: Total number of belief nodes extracted by the LLM for each participant in wave 1 (mean: 9.2) and wave 2 (mean: 8.9), without the maximum constraint of 10 nodes. *Right*: Total number of network connections created by each participant on the canvas in wave 1 (mean: 9.3) and wave 2 (mean: 9.1).

4.3 Reliability of Semantic Content and Connections

Participants produce reliable semantic content over time. We establish this by computing the phi coefficient between sets of topics across waves within participants and between participants (Figure 4A). Higher phi coefficient between sets of topics indicate higher overlap and thus more semantic similarity. Within participants we observe $\bar{\phi}_{within} = 0.48$ and between participants we observe $\bar{\phi}_{between} = 0.18$ for the *selected* BERTopic fit (see Methods Section 3.3.2). This corresponds to a separation of $\Delta\phi = 0.30$. Conventional benchmarks for correlation coefficients (by extension ϕ) classify effects in $[\cdot30, \cdot39]$ as moderate and $\phi \geq 0.50$ as large (Cohen, 2013); across all 10 *candidate* BERTopic fits we observe within-participant overlap in the moderate-to-large range ($\phi_{within} \in [0.32, 0.48]$; see Table S6) while between-participant overlap is small ($\phi_{between} \in [0.09, 0.18]$; see Table S6).

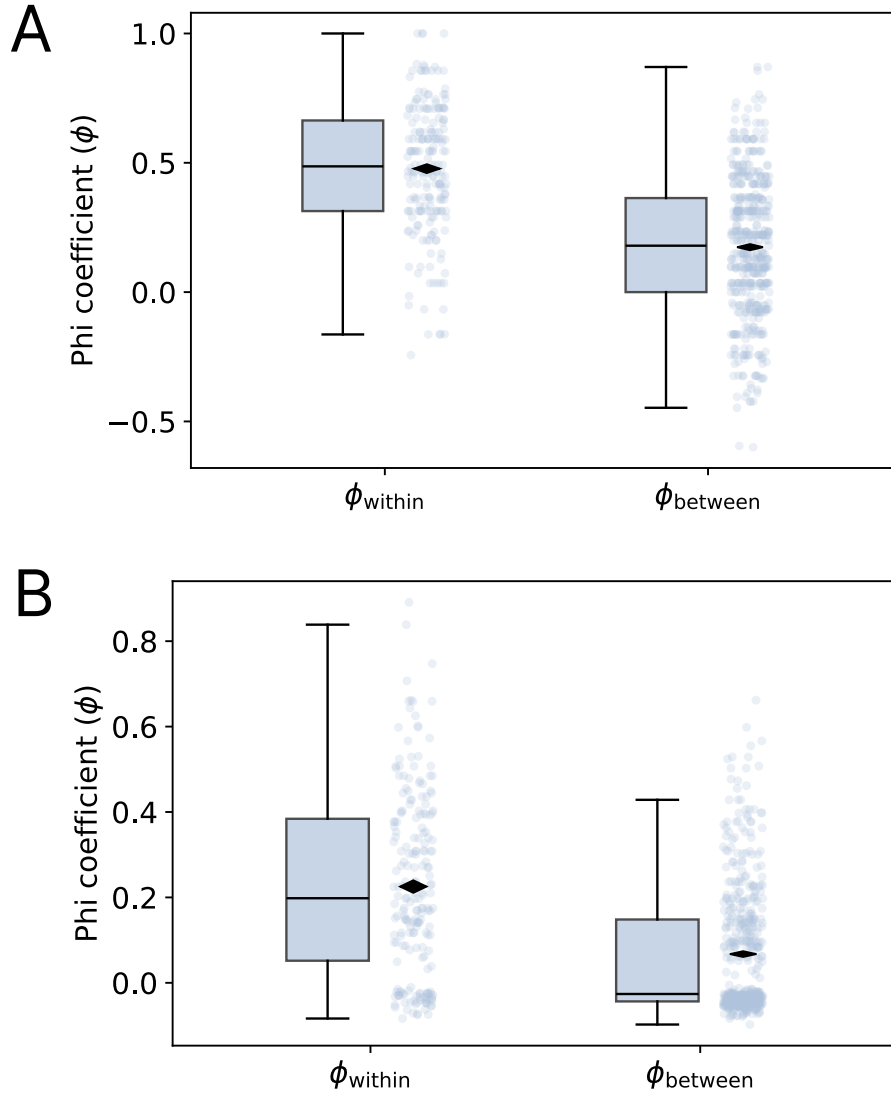


Figure 4: Participants produce semantically reliable content and connections. **A.** Phi-coefficient between binary topic sets ($n = 16$ topics). For ϕ_{within} , this compares topic sets for the same participant across wave 1 and wave 2. For $\phi_{between}$, this compares topic sets for two different participants, one for wave 1 and one for wave 2. For ϕ_{within} , all 210 participants are plotted; for $\phi_{between}$, the figure shows 500 random pairs sampled from the set of possible between-person comparisons across waves. **B.** Phi-coefficient between binary topic connection sets ($n = 136$ possible connections). Applies the same logic as the phi coefficient calculation between topic sets (see Section 3.3.5).

Participants also produce reliable topic connections over time, though with lower reliability than topic presence. We establish this by computing the phi coefficient between binary topic connection vectors across waves within participants and between participants (Fig-

ure 4B). Within participants we observe $\bar{\phi}_{within} = 0.23$ and between participants $\bar{\phi}_{between} = 0.07$ for the *selected* BERTopic fit, corresponding to a separation of $\Delta\phi = 0.16$. By conventional benchmarks (Cohen, 2013), both within- and between-participant values are small ($\phi < 0.30$), which is not surprising given how sparse the topic connection vectors are (see Table S8). Across all 10 *candidate* BERTopic fits, we consistently observe higher within-participant than between-participant phi correlation between topic connection vectors ($\phi_{within} \in [0.11, 0.23]$, $\phi_{between} \in [0.02, 0.07]$; see Table S7).

4.4 Face Validity of Network Representation

Participants rate their own belief networks as more accurate representations of their belief systems than versions with randomly perturbed connections (see Methods Section 3.3.6). Participants rate their own canvas-created network $M = 73$ (SD = 18.4) on a 101-point scale from 0 to 100 (see Figure 5A, left) compared to $M = 43$ (SD = 24.4) for the same network with 50% of the connections randomly flipped (see Figure 5A, right). We observe a similar discrimination for LLM-extracted networks. Participants rate the LLM-extracted network $M = 67$ (SD = 20.9) (see Figure 5B, left), and the same network with 50% randomly flipped connections $M = 38$ (SD = 24.3) (see Figure 5B, right). Taken together, this shows that participants are sensitive to the distinction between “supporting” and “conflicting” network relations, and slightly prefer their own canvas-created networks to those extracted by the LLM.

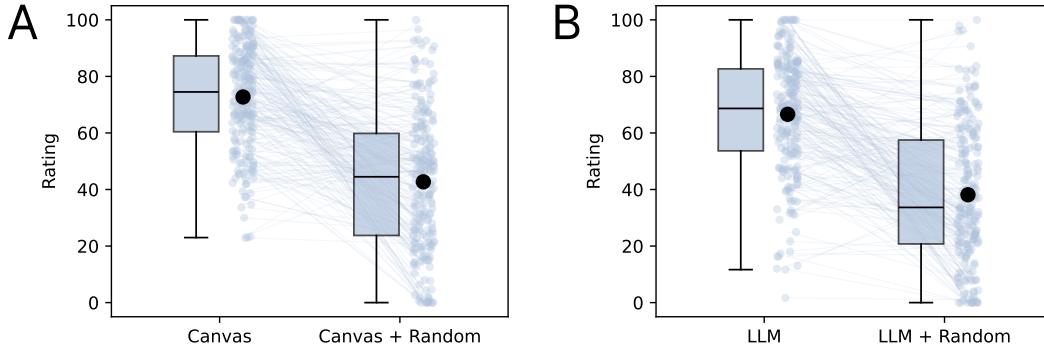


Figure 5: Participants rate their own belief networks as more accurate than versions with randomly perturbed connections. Each dot is a participant-level average over three accuracy ratings of the same network representation, on a 101-point scale anchored at “Not at all” (0) and “Perfectly” (100). Lines connect ratings of the same participant across conditions. The black dot shows the mean. **A.** Ratings of the canvas-created network compared to a version with 50% of connections randomly flipped. **B.** Ratings of the LLM-extracted network compared to a version with 50% of connections randomly flipped.

4.5 Concurrent Validity of Canvas Connections

The connections that participants create using the visual canvas task (see Section 2.3) show reasonable correspondence to connections extracted through other methods. The most direct comparison that we design to assess concurrent validity is the *pairwise evaluation* task (see Section 3.3.7). 58% of all network relationships are coded identically across these two tasks (the diagonal of Figure 6A), with 8% of network relationships “flipped” from either a supporting connection in the visual canvas task to a conflicting connection in the pairwise evaluation task (Figure 6A, top right) or in the other direction (Figure 6A, bottom left). The most common inconsistency involves nodes that participants do not connect in the visual canvas task but do connect in the pairwise evaluation task (27%), showing that when asked directly about how two beliefs relate, participants are more likely to report them as connected in some way than when using the visual canvas task. Restricting the analysis to only cases where network relationships are coded by participants in both tasks (excluding “no connection”) we observe that 84% of network relationships are coded identically.

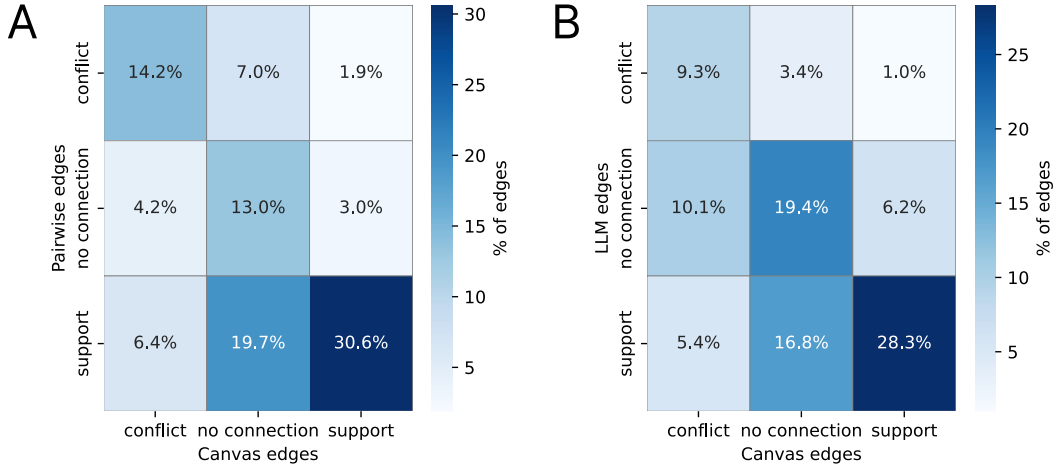


Figure 6: *Concurrent validity*. Canvas-created connections show reasonable correspondence with both pairwise evaluations and LLM predictions. Each cell shows the percentage of all node pairs falling into that combination of categories. Diagonal cells represent agreement between methods. **A)** Concurrent validity between the *visual canvas task* and the *pairwise evaluation task*. **B)** Concurrent validity between the *visual canvas task* and *LLM-extracted connections*.

During the study, we also extract network relations between beliefs using an LLM (see Section 3.3.1). We observe that 57% of network relationships are identical between the visual canvas task and the LLM extraction (the diagonal of Figure 6B), with 6% of network relationships with opposite signs. The most common inconsistency is belief pairs that the participant does not connect but which the LLM assigns a supporting connection (17%), followed by pairs that the participant codes as conflicting but the LLM assigns as no connection (10%). Restricting the analysis to cases where network relationships are coded as either supporting or conflicting in both the canvas task and LLM extraction, we observe that 86% of network relationships are identically coded, slightly higher than between the visual canvas task and the pairwise evaluation task.

4.6 Criterion Validity of Network Structure

Topics that are more interconnected in the belief networks of participants in the first wave are more likely to be present in the second wave (see Section 3.3.8). A theoretical prediction is that topics that are more interconnected with other topics at the individual level should be more likely to persist across waves, consistent with work showing that higher connectivity of

attitude networks is associated with greater stability over time and resistance to change (Dalege et al., 2017, 2025). In line with this theoretically driven hypothesis, we find that topics with a higher degree centrality in the canvas-created networks of participants in wave 1 are more likely to be present in the networks of the same participant in wave 2 (see Figure 7A). In a logistic mixed-effects regression predicting presence of a topic in wave 2 based on its weighted degree in wave 1, with random intercepts for participant and topic, we find a significant positive effect of the weighted degree ($b = 0.13$, $SE = 0.03$, $OR = 1.14$). Random intercept variance was negligible for participants ($SD = 0.00$) but substantial for topics ($SD = 0.66$), indicating that wave 2 topic presence varied more across topics than across participants, as expected. We observe a qualitatively similar pattern when conducting the same analysis using networks based on LLM-extracted connections from participant interviews (see Figure 7B). This pattern holds for all 10 *candidate* BERTopic fits and for both canvas-created connections and LLM-extracted connections (see Figure S12). In Table S9 we show the mean degree and persistence rate for each topic based on the *selected* BERTopic fit.

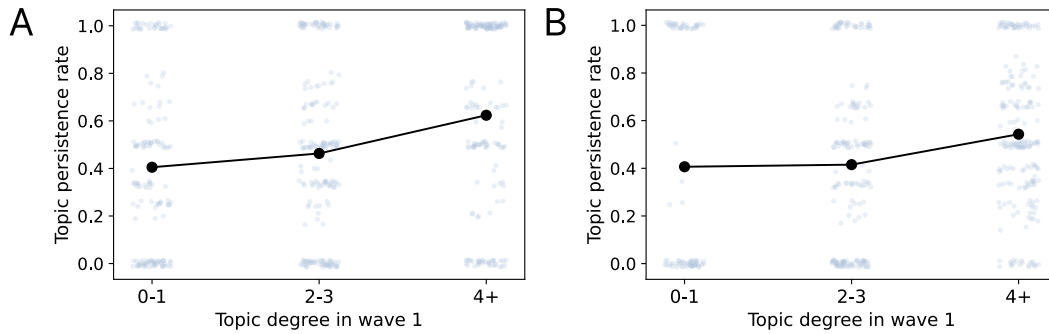


Figure 7: *Criterion validity*. Topics that are more interconnected in wave 1 are more likely to persist into wave 2. **A.** Rate at which a topic is present in wave two conditional on the weighted topic degree of that topic in wave one (binned into 0-1, 2-3, 4+). Raw points are shown for each participant that has at least one topic in a given topic degree bin. Black dots are mean rates calculated across the shown points. **B.** Same as figure A. but based on LLM-extracted connections instead of the connections that participants create on the canvas.

4.7 Additional results

4.7.1 Collective Belief Network. The individual belief networks that we measure with the VN Tool allow us to construct a meaningful network structure at the collective level and we can conceptualize individuals as occupying a subspace of this network. We obtain a collective belief network by aggregating the participant-level networks from the visual canvas task (see Figure 8).

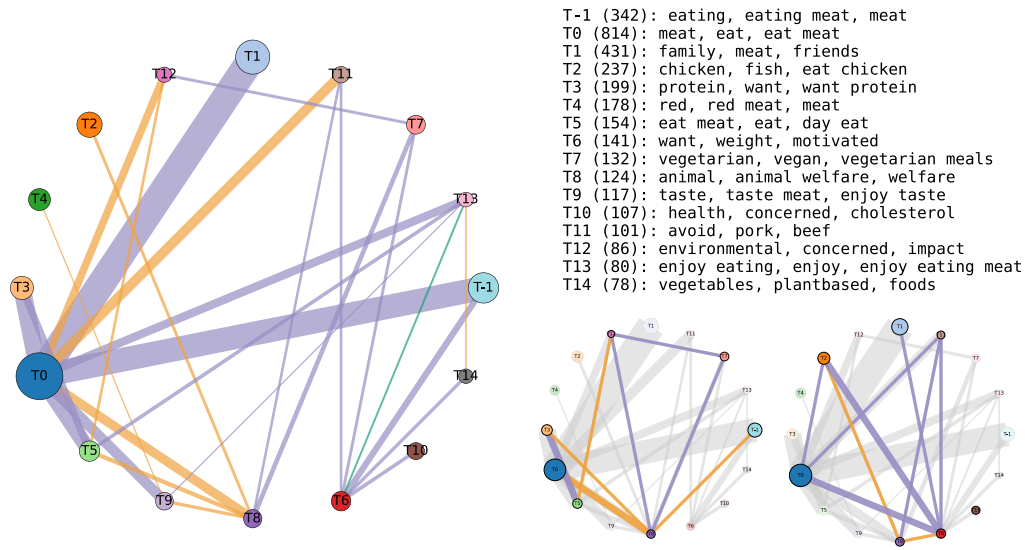


Figure 8: Collective topic network aggregated from participant-level canvas networks (left) and two participant-level topic networks overlaid on the population-level structure (bottom, right). Node size reflects the number of statements assigned to each topic; edge width reflects the number of participant networks containing the connection (k_{ij}). Edges are filtered to include only connections that occur more frequently than expected and with some minimum absolute frequency (see Section S1.9). Purple edges indicate predominantly supporting connections; orange edges indicate predominantly conflicting connections; green edges indicate even split.

Topics -1, 0, and 5 capture personal meat-eating behaviors and personal history, while topic 1 captures social meat-eating behavior. Topics 2 and 4 describe types of meat, splitting broadly into white meat (topic 2; mostly consuming) and red meat (topic 4; mostly limiting). Several topics (3, 6, 10) describe health and nutrition aspects, with the largest (topic 3) being about protein (pro-meat), and topics 6 and 10 mostly capturing health concerns related to meat-eating (e.g., cholesterol). Topics 9 and 13 describe taste and enjoyment associated with meat-eating.

Two topics describe ethical concerns about meat eating; animal welfare (topic 8) and environment (topic 12). Topic 7 captures vegetarian and vegan identity and behaviors, while topic 11 captures avoidance of some meat groups, and topic 14 is about plant-based foods.

These topics are mostly expected given prior research in the domain of meat-eating, which validates our ability to uncover the structure of a domain using a fully bottom-up approach. For instance, animal welfare (topic 8), environmental concerns (topic 12), and health (topics 6, 10) are known to be among the most common motivations to eat less meat (Hopwood et al., 2020), while nutritional considerations (e.g., topic 3), taste and pleasure (topics 9 and 13), and social norms (topic 1) typically support meat-eating (Hopwood et al., 2021). Our model also captures other well-established distinctions, for instance between white meats (topic 2) and red meats (topic 4) (Camilleri et al., 2025; Hayley et al., 2015). We observe that most of the connections that participants create are between behaviors and attitudes; for instance, animal welfare (topic 8) and environmental concerns (topic 12) both conflict with meat-eating behaviors (topic 0). A feature of the VN Tool is that it measures what is spontaneously salient to participants. For instance, while animal welfare (topic 8) and environmental concerns (topic 12) are perceived as conflicting with meat-eating behaviors for the participants that mention them (as expected), we also observe that these ethical concerns do not spontaneously come up in most of the interviews (animal welfare for around 31% of participants and environmental concerns for around 29% of participants; in either wave of the study).

4.7.2 Canvas Training. The network relationships that participants annotate in training mostly obey a simple definition of consistency (see Section 3.4.2), and training improves this. Across all training examples (both wave 1 and wave 2), Overall, 87% of participant-created connections obey the consistency notion that we define. Participants create “conflicting” connections more consistently (91%) than “supporting” ones (84%) during the training phase. The biggest improvement is observed for “supporting” connections which jump from 74% consistent connections for the first training example to 90% for the third training example (see Figure 9). The difference observed in the training consistency between “supporting” and “conflicting” connections is likely due to “supporting” connections being created first, meaning that participants have already received training and feedback before creating their first “conflicting”

connections.

We use an LLM to arrive at the same consistency rule for the network relations that participants create in the visual canvas task (see Section 3.4.2). We observe that consistency for “supporting” connections remains at the same level as during training, while “conflicting” connections are less consistent (see Figure 9). Connections might be psychologically “true” without obeying this consistency requirement, but taken together, our results suggest a consistency ceiling for canvas-created connections of around 85 – 90%.

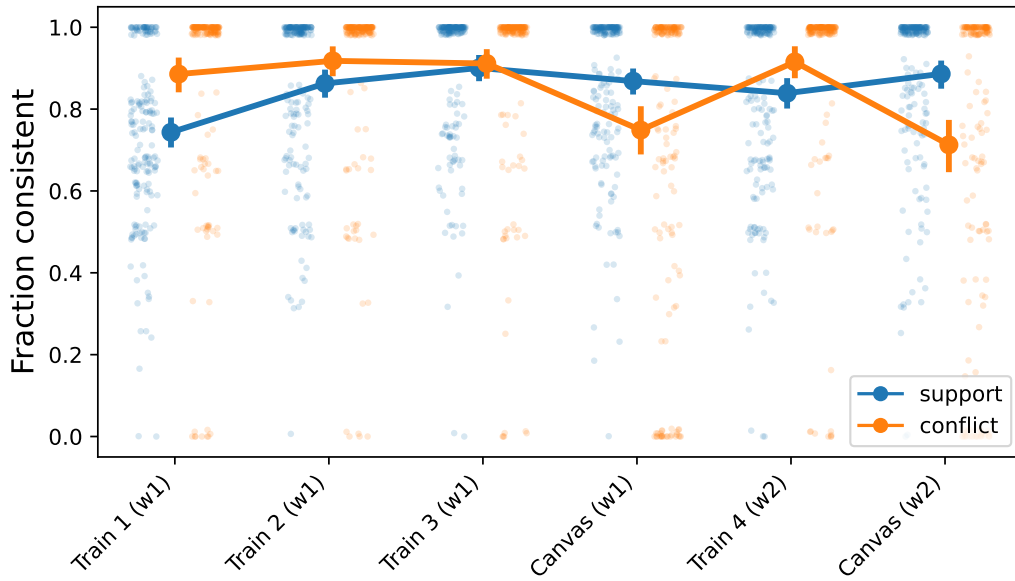


Figure 9: Percent of participant-annotated network relationships that follow a simple consistency rule (see Section 3.4.2). The plot shows average participant-level consistency. We observe a positive effect of training on consistency, especially for “supporting” connections. We observe less consistency for “conflicting” connections in the visual canvas task compared to during training.

5 Discussion

The VN Tool tool elicits individual belief networks using a combination of a LLM-guided semi-structured interview and a visual canvas on which participants annotate relationships between their beliefs extracted from the interviews. The tool does not rely on pre-specified questionnaires that assume the same set of beliefs for all participants, and allows for different relation-

ships at the individual level. Participants report beliefs that are relevant to them within a given domain and annotate how these beliefs are related. We show that basic network properties and semantic content are reliable within participants across waves. We also find support for the validity of the tool. Participants recognize their own canvas-created belief networks when compared to random networks. To a large extent, the belief relations created with the VN Tool match the belief relations indicated through direct pairwise evaluation and LLM extraction. We show that more interconnected beliefs are more likely to persist over time, consistent with the prediction that belief network structure should constrain belief persistence (Dalege et al., 2017, 2025). Finally, the collective belief network that we obtain through aggregating individual belief networks fits to plausible narratives. We obtain this collective representation in a novel and bottom-up way that allows participants to adhere to only part of this collective narrative.

Compared to longitudinal methods for inferring individual belief networks, the VN Tool is less cost-intensive and more suitable for measuring the relations between beliefs that are relatively stable over time because it does not rely on belief change to measure belief relations. Compared to pairwise comparison methods, it places less cognitive demand on participants while still capturing relationships between beliefs. More fundamentally, the VN Tool avoids explicitly asking participants to consider all possible relationships between their beliefs, which might induce reports of previously non-existing connections. In addition, participants do not have to judge each relationship in isolation, but can consider other beliefs when evaluating how beliefs are related.

Tools like the VN Tool presented here could allow testing of individual-level hypotheses about belief dynamics that have previously not been feasible. Core assumptions of belief network theories have not been empirically testable given available methods (Brandt & Vallabha, 2023). For instance, the prediction of “dynamic constraint”—if one belief changes, then closely connected beliefs should come under pressure as well—has only been evaluated using collective belief networks (Turner-Zwinkels & Brandt, 2022). Brandt and Vallabha (2023) has attempted to test theoretical belief network predictions at the individual level using a pairwise conceptual similarity task to measure individual belief networks. While this addresses the gap in the research, their empirical investigation relied on pre-specified and relatively few beliefs

to measure individual belief networks, which could miss beliefs that are individually important (Brandt & Vallabha, 2023).

In addition, most previous research on belief networks infer belief networks on the population level from correlations between participants (Boutyline & Vaisey, 2017; Brandt et al., 2019; DellaPosta et al., 2015). Few previous studies have asked participants to directly report how they perceive beliefs, motivations, or behaviors to be related (Brandt, 2022; Brandt & Morgan, 2022). An advantage of population-based methods is that they can potentially capture implicit associations that might not be available to introspection, but as discussed above this approach also constrains the types of belief networks that we can study. On the other hand, the ability of participants to accurately introspect about how beliefs relate to each other is not well understood outside of conceptual similarity judgment tasks (Brandt, 2022). Our results indicate that when participants are asked about “conflicting” and “supporting” connections they can report these in a relatively reliable way.

There are several avenues for further development of the VN Tool, or similar tools, in the future. One promising avenue is to investigate how the tool can be scaled up to measure much larger belief networks. Currently, the belief networks that we measure with the VN Tool are sparse. This sparsity may reflect genuine features of how participants organize their beliefs, but it may also reflect how participants engage with the task or interpret what constitutes a link. When belief networks are sparse, small differences in the number of reported links can have large effects on network measures, making it difficult to separate belief structure from features of the elicitation process. Furthermore, the notions of “support” and “conflict” between different beliefs could be understood differently by different participants. We do observe that the connections that participants create using the VN Tool and the pairwise comparison are largely consistent, but participants sometimes only report a connection using one of the elicitation methods, and they occasionally report connections in opposing directions. The present study uses a commercial LLM and focuses on meat-eating, but our ongoing research shows that the VN Tool can be successfully implemented with commercial or open-source LLMs, and used to elicit beliefs about different topics.

The major underlying question in the background of the present paper is how best to mea-

sure and compare belief networks over time; both within- and between participants. In this study, we extract belief nodes as open stances (Akash et al., 2025; Velez et al., 2025) which is maximally ideographic and exhaustive. However, this approach introduces the challenge that each set of belief nodes is not directly comparable to any other set, even within the same participant over time. We solve this “mapping problem” using topic modeling (Grootendorst, 2022), but this is imperfect because two belief nodes can be about the same topic but have opposite implications—e.g., “I never eat meat” and “I always eat meat”. One promising avenue is to combine topic modeling with modern LLMs to achieve a better mapping between belief nodes and a shared set of stances (Tamkin et al., 2024). Another possibility is to define a set of pre-specified belief nodes and use an LLM to detect whether each of these is present in any given interview transcript (Bestvater & Monroe, 2023; Burnham, 2025; Wright et al., 2026). Further empirical testing is required to understand which method produces network representations that are most predictive of future belief dynamics. In sum, we have presented a new tool to measure individual belief networks, enabling direct empirical tests of how belief network structure relates to belief persistence and change at the individual level.

Open Practices Statement

Materials and analysis code are available at <https://github.com/victor-m-p/beliefs-narratives-networks>.

Acknowledgments

LLMs were used to revise sentence structure, to develop the VN Tool (in particular the frontend design), to refactor analysis code, and in some cases to provide boilerplate code for analysis and data wrangling.

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S1 Supplementary Information

S1.1 Visual Network Tool Description

In this section, we document one complete run of the Visual Network (VN) Tool survey instrument completed on 2026-02-04 by the lead author. A full video of this completion, along with code to deploy the VN Tool, and supporting scripts, can be found at <https://github.com/victor-m-p/beliefs-narratives-networks>.

S1.1.1 LLM Semi-structured Interview. As described in Section 2.1, the first part of the VN Tool is an 8-round semi-structured interview with an LLM-interviewer. For ethical reasons, we cannot share the raw participant interview transcripts, but we document an interview completed by the lead author on 2026-02-04.

Interview About Meat-Eating

You: Yeah, I mean, it's not really something I navigate moment to moment so much. I feel I've kind of made this decision that I can't buy meat in my grocery store and cook it. I'll occasionally cheat and get something like fish, like salmon, but I've completely cut out other meat. And then when I'm out, for instance, or I don't know, they're grilling at my work or whatever I'm on vacation with my family, I just won't think about it. I'll eat what I want. So this I feel makes my overall impact a lot smaller, but still allows me to indulge sometimes and eat. I mean, I love food, I love trying new things, and sometimes when you go out to eat, it's like it's just the more interesting option still. So this is how I do it now.

Interviewer: I appreciate how thoughtfully you've described your approach and the balance you try to maintain. I'm curious, do the people close to you—like friends, family, or colleagues—have similar eating habits, or do their choices differ from yours?

Transcript preview

[Edit](#)

So, I feel my family is not so relevant, at least anymore, because I live in a different country now. I mean, I grew up eating meat, not crazy amounts, it's just normal to mostly have meat. I'd say friends and colleagues here now are more important. I moved to Vienna around a good year ago now, and my partner here and most of my close friends here are definitely more restrictive than me. Many of them are vegetarian, but some of them just restrict a lot. Some of them even also restrict more, so like won't eat, let's say, eggs, cheese, like other animal products. So that's definitely something that's also influencing me and my habits.

Please skim and correct if needed. If you are happy click "Submit Answer"

 Record Again

 Recording saved (64s)

 Listen back

[Prefer writing?](#)

Interview progress: 4 / 8

Submit Answer

Figure S1: Screenshot of the semi-structured interview about meat-eating with an LLM interviewer. In green a previous utterance by the lead author, in grey the follow-up question by the LLM, and in white (“Transcript preview”) the transcription (Whisper) of the participant answer to the present question.

Interviewer: In this interview, we’re talking about meat-eating habits. Is this something you’ve thought much about, or not really?

Participant: It’s something I’ve thought quite a bit about, I mean now I also read about it and do research about it, but even before that it was, I mean it’s come up a few times, especially when I’ve moved in with people, let’s say that are vegetarian or just restrict meat, when I’ve gotten to know people very well that have different habits, yes it’s definitely something that’s been coming up a few times and also where I’ve changed my own habits a bit and have read about factory farming practices, etc. Yes I would say it’s something that is like a salient topic in my circles and is something I’m also navigating myself.

Interviewer: Thanks for sharing that—it's interesting to hear how your habits and thoughts have evolved, especially as you've learned more and spent time with people who have different approaches. Could you tell me a bit about your own current meat-eating habits?

Participant: Yeah, so my own current meat-eating habits are, I mean, I still eat meat, right, but I think what I settled on for now is just that I don't eat meat regularly, I don't buy meat and cook it myself, but I'll allow myself to eat meat if it's offered, if I'm out, if it's interesting, but I mostly just try to not buy cheap meat, where I know that the animals have suffered, so that's kind of the compromise I've arrived on for now.

Interviewer: Thanks for explaining the compromise you've found with your meat-eating—it's interesting how you balance your choices based on circumstances and concerns about animal welfare. Could you share a bit more about what motivates you personally to eat meat on some occasions, or to avoid it on others?

Participant: Yeah, I mean, it's not really something I navigate moment to moment so much. I feel I've kind of made this decision that I can't buy meat in my grocery store and cook it. I'll occasionally cheat and get something like fish, like salmon, but I've completely cut out other meat. And then when I'm out, for instance, or I don't know, they're grilling at my work or whatever I'm on vacation with my family, I just won't think about it. I'll eat what I want. So this I feel makes my overall impact a lot smaller, but still allows me to indulge sometimes and eat. I mean, I love food, I love trying new things, and sometimes when you go out to eat, it's like it's just the more interesting option still. So this is how I do it now.

Interviewer: I appreciate how thoughtfully you've described your approach and the balance you try to maintain. I'm curious, do the people close to you—like friends, family, or colleagues—have similar eating habits, or do their choices differ from yours?

Participant: So, I feel my family is not so relevant, at least anymore, because I live in a different country now. I mean, I grew up eating meat, not crazy amounts, it's

just normal to mostly have meat. I'd say friends and colleagues here now are more important. I moved to Vienna around a good year ago now, and my partner here and most of my close friends here are definitely more restrictive than me. Many of them are vegetarian, but some of them just restrict a lot. Some of them even also restrict more, so like won't eat, let's say, eggs, cheese, like other animal products. So that's definitely something that's also influencing me and my habits.

Interviewer: Thanks for explaining how your friends and partner's habits influence you. I'm curious, do you have a sense of what motivates them to be more restrictive with meat or other animal products?

Participant: Yeah, I think it differs a bit. I do think, I mean, some of them made this decision many years ago, right, to just cut meat out completely. Yeah, I mean, I think some of them, many of my close friends here are very concerned about climate change and environmental impact. I think also some of them are very concerned about animal welfare, which I think, I mean, that's definitely my bigger concern with this. Then I think there's like smaller differences, right? I think I'm mostly concerned about animal suffering. I won't feel so bad about eating meat if I know that the animal has lived a decent life. But I think some of them are just fundamentally against, let's say, any kind of farming for meat. I guess that's sort of philosophically a little bit more, I don't know, Kantian, if I'm like a utilitarian, but maybe that's like overthinking it. I mean, I also just think it's kind of a part of a left-wing identity. Many of my friends here are more active on the left wing, also organizing, protesting, doing different things. And I think being a vegetarian is definitely like kind of that package in a way.

Interviewer: Thanks for sharing those reflections about your friends' and your own motivations—it's really interesting to hear how different concerns like animal welfare, the environment, or philosophical beliefs factor in. You mentioned that being vegetarian can also be part of a broader identity or community for some people—could you say a bit more about how you see that playing out among your friends or in your social circles?

Participant: Yeah, I'm not sure how much more I can elaborate on that. I just think that most people in my circle, me included, do hold the belief that eating less meat is the more moral and ethical choice for lots of reasons. It just is true that it has much less environmental impact, and it's also true that many animals suffer greatly in factory farming. I actually don't think that my friends and my partner are judgmental. I think they're pretty chill, and most of them also will occasionally cheat and try meat, maybe not all of them. So I think they are also just showing by example that this is a choice you can make, and yeah, I think that is also just inspiring me to reconsider my own choice and how I go about this.

Interviewer: Thanks for sharing how your friends and partner influence you and how shared beliefs shape your group's approach—it's interesting to hear about the positive, non-judgmental way this plays out. Looking back at your journey, has there been a particular moment or experience that really shifted the way you think about eating meat?

Participant: Good question. I mean, it's a little bit difficult to pinpoint one moment. I think there's been a couple of different things. I mean, I do think as many others, one thing that has shocked me has been seeing clips and videos from factory farms. I think for many people, and rightly so, this is a shocking experience. I also a few years ago read Peter Singer's *Animal Liberation*, which sort of documents how terrible factory farming is. And so the scale of suffering involved from a very utilitarian perspective, but it was still a very impactful book for me to read. I mean, to be fair, I've also been convinced for many years that I should eat less meat, and I've still maintained eating meat, right, because of various things. Because I'm not entirely self-consistent, I enjoy eating it, etc. So I think maybe what has more driven me to start changing has been, let's say, the first time a few years ago when I moved in to a collective with vegetarians. That's definitely when my day-to-day habits started changing and maybe more coming into alignment with my beliefs.

Interviewer: Thank you for sharing so much about your experiences and reflections—it's been really insightful. Before we wrap up, is there anything else about your

thoughts, experiences, or journey around eating meat that you'd like to mention that we haven't covered yet?

Participant: Yeah, I mean, I guess, I mean, one thing we haven't talked so much about is why I still do eat meat sometimes. I think we've talked a lot about all of my motivations, both social and moral, to cut down on meat. I do think there is, I mean, I greatly respect the people that have made a more consistent sort of choice than I have, to really just cut out meat completely. And that might also be a little bit easier than having to evaluate from moment to moment whether you should allow yourself, but I also think that a lot of people could cut down 50, 60, 80% of their meat intake without experiencing much, quote-unquote, disruption. I definitely feel that so far I haven't made any difficult decisions. It's very easy for me, maybe not for everyone, I also respect that, but it's easy for me to cook great vegetarian food myself and then I still allow myself to experience sometimes great tasting meat when I'm out, and to not turn down food, to eat whatever is offered. So right now I'm actually pretty happy with having reduced so much, but still sort of, yeah, not having restricted myself so much, but let's see.

After completing the LLM-guided semi-structured interview, participants answer a few questions that help us assess the quality of the interview and participant experience (see Figure 1A). All of the questions below are answered on a 101-point scale from 0 to 100 with dynamically changing anchors. All questions are taken from OpenResearch (2025b). We observe consistently high ratings for all items (see Figure S10B; notice that the last item is reverse-coded, so a low rating is desirable).

1. "How was your overall experience with the conversational survey?" (Terrible, Not good, Average / Neutral, Good, Excellent).
2. "The questions in the interview were relevant" (Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree).
3. "It was easy to express my thoughts in the chat format" (Strongly disagree, . . . , Strongly agree).

4. “I felt comfortable enough during the conversation to honestly describe my habits and motivations” (Strongly disagree, . . . , Strongly agree).
5. “Interacting with the AI model felt creepy or intrusive” (Strongly disagree, . . . , Strongly agree).

S1.1.2 Extraction and Rating of Belief Statements. As described in Section 2.2, the LLM extracts up to 10 belief statements based on the participant interview transcript. Participants rate how *accurate* and *relevant* each statement is, including two attention checks. Based on the interview with the lead author on 2026-02-04 the following 10 statements were extracted:

1. I do not buy meat to cook
2. I eat meat when eating out
3. I avoid cheap meat
4. I love trying new foods
5. I am concerned about animal suffering
6. I am concerned about environmental impact
7. My social circle restricts animal products
8. Social contacts are concerned about animal welfare
9. Videos of factory farming shocked me
10. Reading Animal Liberation influenced me

And presented in random order along with the two attention checks: (1) “I eat beans to get enough cholesterol”, and (2) “My friends often go out to eat seahorse”.

How accurate are these statements?

The following are short AI-extracted summaries that are extracted based on the interview we just had. There are also attention checks on this page. We are interested in learning how accurately you feel that each of these **accurately reflect what you said in the interview**.

Your task: Move each slider to rate **how accurately** each statement reflects what you said in the interview.

Scale: 0 (Not accurate at all) to 100 (Very accurate).

Focus only on how accurate each statement is. You will get the chance to rate how relevant each statement is later.

Show transcript

I eat beans to get enough cholesterol

Not accurate at all
Very accurate

Selected: Not accurate at all — 0 / 100

I am concerned about environmental impact

Not accurate at all
Very accurate

Selected: Very accurate — 96 / 100

My friends often go out to eat seahorse

Not accurate at all
Very accurate

Selected: Not accurate at all — 0 / 100

I am concerned about animal suffering

Not accurate at all
Very accurate

Selected: Very accurate — 100 / 100

Figure S2: Screenshot showing the page where participants rate the *accuracy* of belief statements extracted by the LLM. In this screenshot we see ratings of two attention checks: “I eat beans to get enough cholesterol” and “My friends often go out to eat seahorse” along with two LLM-extracted summaries based on the interview transcript (the participant scrolls down to rate all statements on one page). The interface for the *relevance* ratings of belief statements is identical.

We observe consistently high ratings on both *accuracy* and *relevance* for LLM-extracted belief statements across both waves, and consistently low ratings on both *accuracy* and *relevance* for the two attention checks (see Figure S10B). The specific questions answered are:

1. “Rate how **accurately** each statement reflects what you said in the interview” (101-point slider from 0 to 100 with 7-point Likert annotations, anchored at “Not accurate at all”, “Very accurate”)
2. “Rate how **relevant** each statement is for your personal meat-eating habits” (101-point slider from 0 to 100 with 7-point Likert annotations, anchored at “Not relevant at all”, “Very relevant”)

“Very relevant”).

S1.1.3 Visual Canvas Task. As described in Section 2.3, the visual canvas task consists of 3 parts; spatial arrangement, “supporting” connections, and “conflicting” connections. Because the lead author rated all 10 belief statements as at least “Accurate” (60 or above on a 101-point scale) all 10 statements are used here.

Which things go together?

Your task: Place things that go together closer to each other. Do this based on how things go together for you personally.

Note: You can view your interview transcript by clicking the button below.

✅ AI analysis complete. You can continue.

[Show transcript](#)

Drag into the box →

I do not buy meat to cook

I am concerned about environmental impact

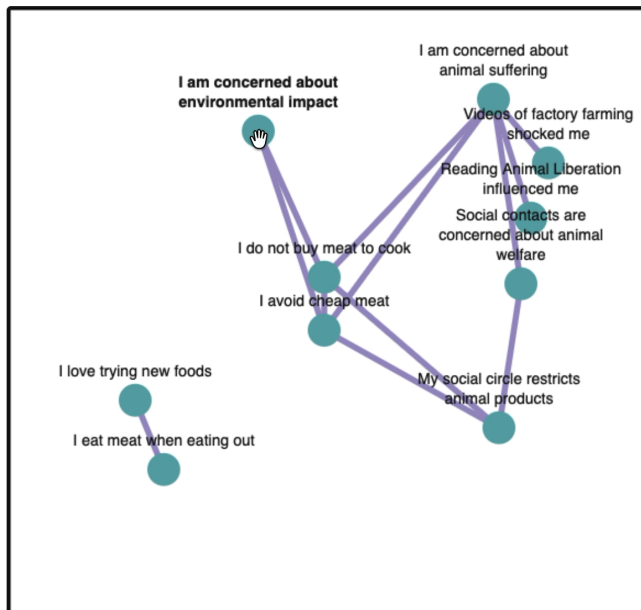
Figure S3: Screenshot showing the first part of the visual canvas task which is spatial arrangement of belief statements. Initially, the belief statements appear in random order on the right (“Drag into the box”). Participants can proceed when they have dragged all belief statements into the canvas on the left. The extraction of LLM edges (see Section 3.3.1) is executed in the background while participants complete this task.

Which things support each other?

You already placed things based on how they go together for you personally. Now please connect **blue circles** that **support each other** for you personally. By support each other we mean that one thing reinforces or encourages another. For now, focus only on **supporting** relations. We will ask you about any conflicting relationships on the next page.

Note: You can view your interview transcript by clicking the button below.

Show transcript



How to connect circles

1. Click a **blue circle** to select it.
2. Click another **blue circle** to create a **supporting** connection.
3. Click the same pair again to remove the connection.

You can still move circles around the canvas if this helps you organize things.

Figure S4: Screenshot showing the second part of the visual canvas task, which involves creating “supporting” connections. Participants click on one belief statement to activate it, and click on another belief statement to complete the connection. Participants can continue to arrange (move) their belief statements.

Which things are in conflict with each other?

On the previous page you connected things that support each other. We are also interested in things that are in conflict with each other. Please connect **blue circles** that **conflict with each other** for you personally. By conflict we mean that one thing contradicts or undermines another. Focus only on **conflicting** relations.

Note: You can view your interview transcript by clicking the button below.

[Show transcript](#)

How to connect circles

1. Click a **blue circle** to select it.
2. Click another **blue circle** to create a **conflict** connection.
3. Click the same pair again to remove the connection.

You can still move circles if this helps you organize things.

Figure S5: Screenshot showing the third part of the visual canvas task, which involves creating “conflicting” connections. Participants click on one belief statement to activate it, and click on another belief statement to complete the connection. Participants can continue to arrange (move) their belief statements.

S1.1.4 Canvas Training. As described in Section 3.2.1 participants complete a training exercise before the visual canvas task in both wave 1 (3 training examples) and in wave 2 (1 training example). The training task follows the structure and the visual layout of the canvas task described above. First, participants visually arrange statements, then they create “supporting” connections, and then they create “conflicting” connections.

Training exercise

Please read the short story below carefully. A few short statements were extracted from this text. In the next steps you will practice placing these statements on a screen and creating supporting and conflicting connections.

Vignette

Alex decided to learn Spanish to improve job prospects and to be able to speak with locals while traveling. To practice speaking, Alex occasionally attends a conversational meetup. Some of Alex's colleagues are also learning Spanish and attending the meetup, and this motivates Alex to go as well. However, Alex feels very embarrassed speaking out loud, and this sometimes makes Alex avoid the meetup. To remain consistent, Alex tries to practice Spanish every day using a language-learning app. Alex has a busy job, and when it gets hectic Alex skips the daily practice.

Extracted statements

The following statements were extracted based on the short story above. You will use these in the next practice step.

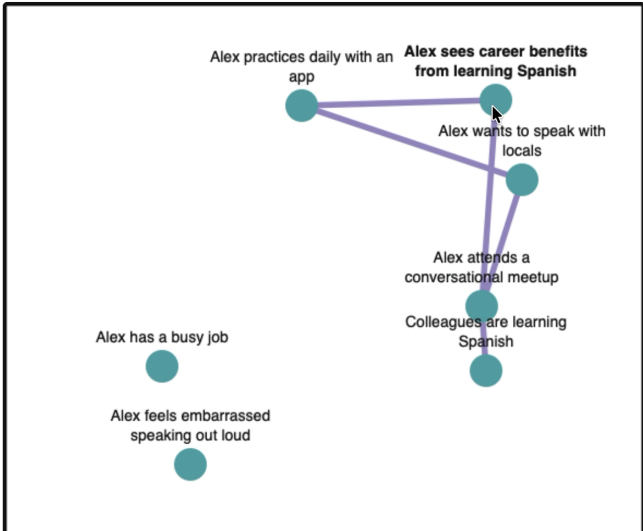
1. Alex sees career benefits from learning Spanish
2. Alex wants to speak with locals
3. Alex practices daily with an app
4. Alex has a busy job
5. Alex attends a conversational meetup
6. Colleagues are learning Spanish
7. Alex feels embarrassed speaking out loud

Figure S6: Screenshot showing an information screen with a training vignette and the pre-defined belief statements associated with it. This is shown to participants before they proceed to the spatial arrangement task for each training example.

Which things support each other?

You already placed things based on how they go together for Alex. Now please connect **blue circles** that **support each other** for Alex personally. By support each other we mean that one thing reinforces or encourages another. For now, focus only on **supporting** connections. We will ask you about conflicting connections on the next page.

Here is the vignette for your reference: Alex decided to learn Spanish to improve job prospects and to be able to speak with locals while traveling. To practice speaking, Alex occasionally attends a conversational meetup. Some of Alex's colleagues are also learning Spanish and attending the meetup, and this motivates Alex to go as well. However, Alex feels very embarrassed speaking out loud, and this sometimes makes Alex avoid the meetup. To remain consistent, Alex tries to practice Spanish every day using a language-learning app. Alex has a busy job, and when it gets hectic Alex skips the daily practice.



How to connect circles

1. Click a **blue circle** to select it.
2. Click another **blue circle** to create a **supporting** connection.
3. Click the same pair again to remove the connection.

You can still move circles around the canvas if this helps you organize things.

Figure S7: Screenshot showing the creation of “supporting” connections for the “Alex” training scenario. The page works identically to the page where participants create supporting connections between their own meat-eating belief statements. The page for “conflicting” connections is visually identical.

We design 3 training examples (see below), which all take the same basic form. A text with some goal is provided (e.g., learning Spanish) and 7 belief statements are provided, encompassing both the goal, motivations and challenges. In wave 1 each participant completes all 3 training examples in random order and in wave 2 each participant completes 1 training example sampled randomly. We provide live feedback to participants and calculate a measure of consistency for participant codings during the training phase (see Section 3.4.2).

Vignette 1 (“Alex”)

Alex decided to learn Spanish to improve job prospects and to be able to speak with locals while traveling. To practice speaking, Alex occasionally attends a conversational meetup. Some of

Alex's colleagues are also learning Spanish and attending the meetup, and this motivates Alex to go as well. However, Alex feels very embarrassed speaking out loud, and this sometimes makes Alex avoid the meetup. To remain consistent, Alex tries to practice Spanish every day using a language-learning app. Alex has a busy job, and when it gets hectic Alex skips the daily practice.

1. Alex sees career benefits from learning Spanish
2. Alex wants to speak with locals
3. Alex practices daily with an app
4. Alex has a busy job
5. Alex attends a conversational meetup
6. Colleagues are learning Spanish
7. Alex feels embarrassed speaking out loud

Vignette 2 ("Jordan")

Jordan wants to improve their physical fitness and to have more energy during the day. A friend has invited Jordan to join their gym sessions, and the gym is located on Jordan's way home from work, which makes it convenient to go. Jordan plans to go to the gym three times a week. However, Jordan often feels exhausted after work and feels self-conscious exercising in front of others, and these feelings sometimes lead Jordan to skip the gym.

1. Jordan wants to improve their physical fitness
2. Jordan wants to have more energy during the day
3. Jordan plans to go to the gym three times a week
4. Jordan often feels exhausted after work
5. A friend has invited Jordan to join their gym sessions
6. The gym is located on Jordan's way home from work

7. Jordan feels self-conscious exercising in front of others

Vignette 3 (“Riley”):

Riley wants to get more involved in the local community and registered for a weekly volunteer shift. A close friend also volunteers there, which makes Riley feel welcome. The community center is located close to Riley’s home, so getting there is usually easy. However, Riley sometimes needs to take care of a younger relative on short notice, which makes it difficult to attend the volunteer shift consistently. Riley tries to plan ahead each week, using reminders and a shared calendar, but unexpected caregiving needs sometimes interfere.

1. Riley wants to contribute to the local community
2. Riley registered for a weekly volunteer shift
3. A close friend also volunteers there
4. The community center is close to Riley’s home
5. Riley sometimes needs to take care of a younger relative on short notice
6. Riley tries to plan ahead using reminders and a shared calendar
7. Unexpected caregiving needs sometimes interfere with Riley’s plans

S1.1.5 Pairwise Evaluation. We implement a pairwise evaluation task (see Section 3.2.2) to assess the concurrent validity of the visual canvas task. We administer the pairwise evaluation task in the second wave. The pairwise interview alternates between an open question about how two belief statements relate for the participant, and a specific question which forces the participant to choose a classification (see Section 3.2.2). The design for this page is shown in Figure S8.

Interview about how things relate for you

Thank you for drawing your own network of how meat-eating motivations are connected to each other for you. To better understand how things relate for you, we would love to hear you describe a few in your own words. On this page an AI will conduct a brief interview with you. Please record your responses below, like you did in the first interview.

Current pair: I am concerned about animal suffering — Reading Animal Liberation influenced me

Interviewer: Two things that came up in the interview were your concern about animal suffering and the influence of reading Animal Liberation. Are these two things connected in some way for you? And if yes, how?

You: Yeah, I mean, I think I was already concerned about animal suffering, but definitely reading Animal Liberation by Peter Singer was one thing that made it a bigger topic for me, I guess. Yeah, I would say reading Animal Liberation definitely influenced my views on factory farming and animal suffering.

Interviewer: Thanks for clarifying how reading Animal Liberation influenced your thinking about animal suffering. If you had to choose, how would you describe the way these two things relate overall?

They support each other

They conflict with each other

They are not clearly connected

Interview progress

Submit Answer

Figure S8: Screenshot showing the design for the pairwise comparison task. The LLM interviewer first asks an open question about how two belief statements relate (grey textbox at the top) and the participant answers (green textbox in the middle). The LLM interviewer then asks the participant to classify this as either support, conflict, or no connection (grey textbox at the bottom) and the participant decides by clicking and submitting their answer. This structure loops for up to 9 pairs of belief statements.

S1.1.6 Network Comparison. We implement a network comparison task (see Section 3.2.3) to establish face validity (see Section 4.4). This task is only administered in the second wave. Each network comparison screen shows two network representations that the participant rates based on how accurately they reflect their beliefs in the domain of meat-eating (see Figure S9).

Which network best describes your meat-eating influences?

Thank you for completing the interview on how things relate. We will now present you with pairs of networks that might describe how different things relate to each other for you. Please take a careful look at the two networks below and rate how well they describe your meat-eating influences. Only the connections differ, so focus on comparing these. You will compare 6 pairs of networks in total.

Your task: Rate how well each network captures how things relate for you. Then pick the one you think is more accurate overall and briefly explain why you prefer it over the other one.

Network 1

■ Supporting ■ Conflicting

How well does this describe your meat-eating influences?

Network 2

■ Supporting ■ Conflicting

How well does this describe your meat-eating influences?

Figure S9: Screenshot showing the network comparison task design. On the left, the actual network of “conflicting” and “supporting” connections created by the lead author using the VN Tool. On the right, the same belief statements but 50% of the connections with the opposite direction.

S1.2 Questionnaires

In both waves of the study, we ask participants questions about their meat-eating, and whether they feel personal or social dissonance (see Table S1). We do not report analyses based on the scale responses to these questions.

Table S1: Meat-eating frequency and dissonance items.

Item	Scale
<i>Frequency (1–100; Never ↔ Every day)</i>	
1 How often do you eat any meat in an average week?	Present
2 Try to recall your meat-eating habits 5 years ago. How often did you eat any meat in an average week back then?	Past
3 Now imagine 5 years from now. How often do you think you'll eat any meat in an average week?	Future
<i>Dissonance (0–100; Strongly disagree ↔ Strongly agree)</i>	
4 I experience no conflict at all between my personal beliefs/values and my meat-eating behavior.	Personal
5 I experience no conflict at all between my social environment and my meat-eating behavior.	Social

In the second wave of the study, we administer the VEMI (Hopwood et al., 2020) and MEMI (Hopwood et al., 2021) questionnaires on meat-eating. We use the original items and endpoint anchors, but replace the original 7-point Likert scales with a 101-point slider (0-100). For the MEMI questionnaire, the slider is anchored at “Least important” (0), “Moderately important” (50), and “Most important” (100). For the VEMI questionnaire, the slider is anchored at “Not important” (0), “Moderately important” (50), and “Very important” (100). We do not report analyses based on the VEMI and the MEMI questionnaires in this paper. We use the same instruction text as in the original VEMI and MEMI papers.

S1.2.1 Vegetarian Motives Inventory (VEMI). We use the VEMI items (Table S2) as they appear in Hopwood et al. (2020).

Table S2: Vegetarian Motives Inventory (VEMI) items and subscales.

Item	Subscale
1 I want to be healthy	Health (H)
2 Plant-based diets are better for the environment	Environment (E)
3 Animals do not have to suffer	Animal rights (A)
4 Animals' rights are respected	Animal rights (A)
5 I want to live a long time	Health (H)
6 Plant-based diets are more sustainable	Environment (E)
7 I care about my body	Health (H)
8 Eating meat is bad for the planet	Environment (E)
9 Animal rights are important to me	Animal rights (A)
10 Plant-based diets are environmentally-friendly	Environment (E)
11 It does not seem right to exploit animals	Animal rights (A)
12 Plants have less of an impact on the environment than animal products	Environment (E)
13 I am concerned about animal rights	Animal rights (A)
14 My health is important to me	Health (H)
15 I don't want animals to suffer	Animal rights (A)

S1.2.2 The Motives to Eat Meat Inventory (MEMI). We use the MEMI items (Table S3) as they appear in Hopwood et al. (2021).

Table S3: Motives to Eat Meat Inventory (MEMI) items and subscales.

Item	Subscale
1 It goes against nature to eat only plants.	Natural
2 Our bodies need the protein.	Necessary
3 I want to fit in.	Normal
4 It is delicious.	Nice
5 It makes people strong and vigorous.	Necessary
6 I don't want other people to be uncomfortable.	Normal
7 It is in all of the best tasting food.	Nice
8 It could be unnatural not to eat meat.	Natural
9 It is necessary for good health.	Necessary
10 It is just one of the things people do.	Normal
11 It gives me pleasure.	Nice
12 I want to be sure I get all of the vitamins and minerals I need.	Necessary
13 Everybody does it.	Normal
14 It has good flavor.	Nice
15 It gives me strength and endurance.	Necessary
16 I don't want to stand out.	Normal
17 Meals without it don't taste good.	Nice
18 It is human nature to eat meat.	Natural
19 Eating meat is part of our biology.	Natural

S1.3 Evaluation of LLM Interview and Statements

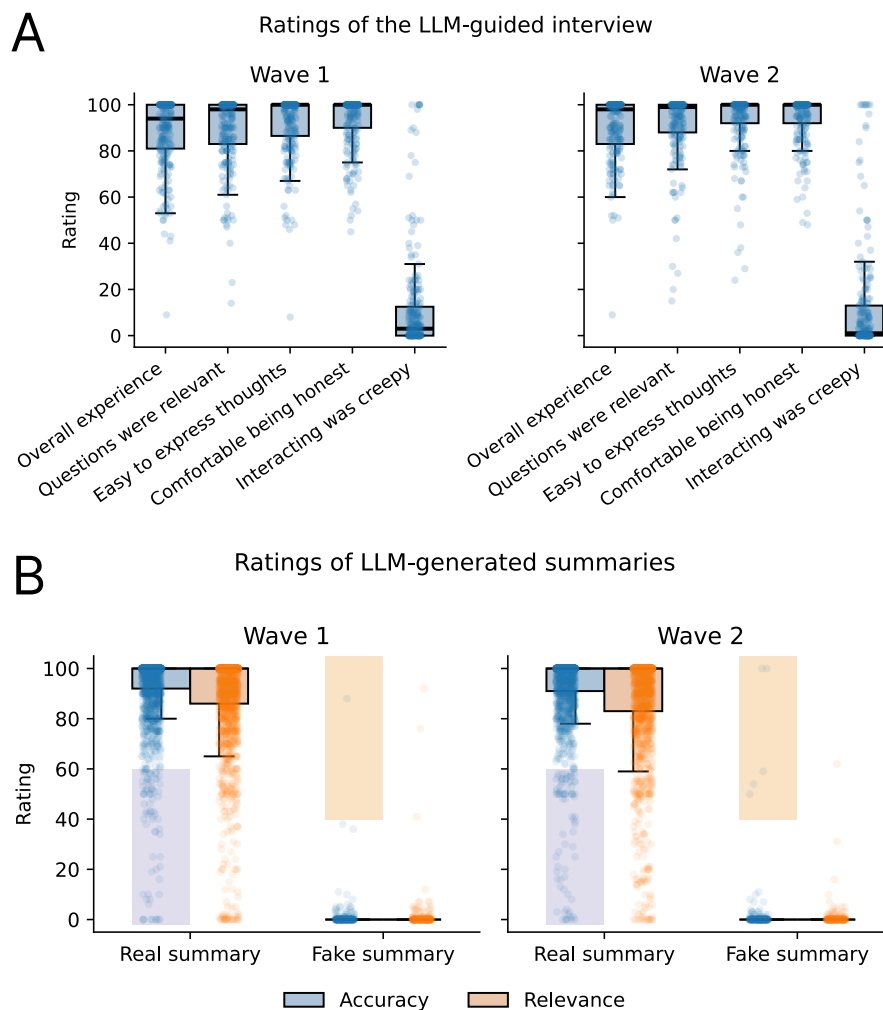


Figure S10: Participants rate their interactions with the LLM positively across both waves. **A.** Participant ratings of the semi-structured LLM interview across five dimensions (see Section S1.1.1) for wave 1 ($n = 247$) and wave 2 ($n = 217$). **B.** Participant ratings of the accuracy and relevance of LLM-extracted belief statements for wave 1 ($n = 247$) and wave 2 ($n = 217$). LLM-extracted belief statements are rated highly on both *accuracy* and *relevance*, while fake summaries (attention checks) are rated as inaccurate and irrelevant. Statements rated below “Accurate” (60 or below on the 100-point scale) are filtered out (shown in purple). Participants that rate the most obvious attention check as “Somewhat accurate” (40 or above on a 100-point scale) are excluded from further analysis (shown in orange).

S1.4 Canvas Proximity and Connection Types

We investigate whether the spatial arrangement of belief nodes on the canvas is predictive of the connections that participants create between their beliefs. For each belief network created on the canvas ($n = 420$ across participants and waves), we compute the Euclidean pixel distance between all pairs of belief nodes and normalize these distances within each network using min-max scaling, such that the closest pair has a distance of 0 and the most distant pair has a distance of 1. We assign each pair to equal-width bins spanning $[0, 1]$ in intervals of 0.1. For each participant and bin we compute the fraction of pairs in that bin connected by a *supporting* edge and the fraction connected by a *conflicting* edge. We calculate the mean of these participant-level fractions across all participants with at least one pair in the bin.

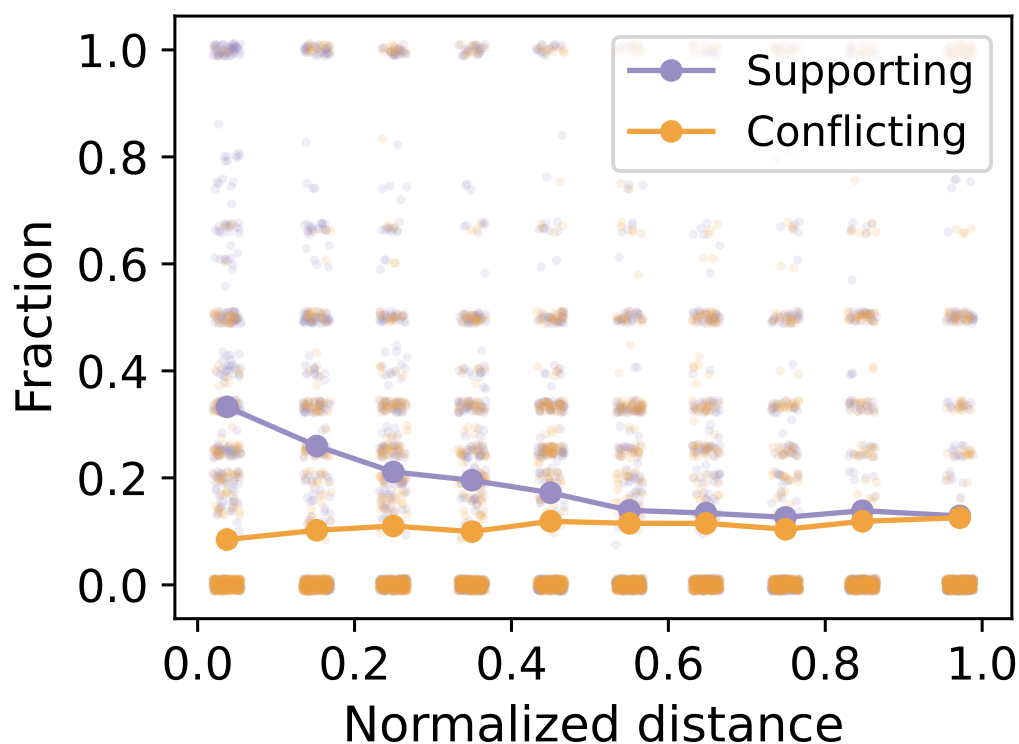


Figure S11: Beliefs that are placed closer to each other on the canvas are more likely to be connected by a supporting edge. For each of the 420 total belief networks (both waves), belief-pair distances are normalized to $[0, 1]$ and grouped into ten bins. Dots show participant-level fractions connected by supporting (purple) or conflicting (orange) edges; solid dots show bin means.

S1.5 Reliability of Network Structure

In Results Section 4.2 we report the number of proposed belief statements (LLM, no limit) and the number of canvas-created edges. Table S4 additionally shows the correlation for the number of interview words spoken or typed by participants (full interview), the number of proposed belief nodes by the LLM during the experiment (LLM, live), the number of participant-accepted nodes (on canvas), and the reliability of canvas edges split by type (supporting, conflicting).

Table S4: Descriptive statistics and reliability

Measure	Mean W1	SD W1	Mean W2	SD W2	r
Interview words	456.11	302.24	449.81	331.18	0.75
Proposed nodes (LLM, no limit)	9.24	2.83	8.90	3.09	0.38
Proposed nodes (LLM, live)	8.31	1.37	8.16	1.53	0.21
Accepted nodes (on canvas)	8.00	1.49	7.82	1.62	0.17
Canvas edges	9.33	4.59	9.07	4.44	0.54
Supporting edges	6.01	3.21	5.61	2.62	0.48
Conflicting edges	3.32	2.45	3.45	2.48	0.42

S1.6 BERTopic Table

The *selected* BERTopic fit (see Methods Section 3.3.2) provides a mapping between all 3458 belief statements (*nodes*: extracted from 2×210 interviews) and 16 topics (including the outlier topic). Below, we present an overview table for this *selected* model (09_all-MiniLM-L6-v2). We observe reasonable separation of the target topics: “family eats meat” (topics 1), “wanting protein” (topics 3), “animal welfare concern” (topic 8) and “environmental concern” (topic 12). It is expected that the outlier topic (topic -1) and the first topic (topic 0) are less coherent than other topics. For instance, while topic 0 mostly captures descriptions of meat-eating behavior, we see some noisy assignments (e.g., BERTopic’s Representative Docs cover both health concerns and meat reduction).

Topic	n	Keywords	Representative Docs
-1	356	OUTLIER	I grew up eating meat, I grew up eating meat at meals, I was raised on a farm
0	871	meat, eat, eat meat, meals, meats	I am concerned about soy and estrogen, I sometimes think about health risks of meat, I aim to reduce meat consumption
1	446	family, meat, friends, social, meat family	Most friends eat a lot of meat, Friends and family eat less meat and pork, Most family and friends eat meat
2	243	chicken, fish, eat chicken, eat, turkey	I mostly eat chicken, I eat beef and chicken regularly, Beef and chicken are my most eaten meats
3	203	protein, want, want protein, meat protein, source	I believe meat provides necessary protein, I want to get enough protein, I feel sick without protein
4	186	red, red meat, meat, eat red, eat red meat	Red meat is traditional in my culture, Red meat causes health complications for me, I feel better eating less red meat
5	159	eat meat, eat, day eat, day, times	I eat meat three days a week, I eat meat two or three times weekly, I eat meat four to five nights weekly
6	146	want, weight, motivated, diet, healthy	I want to avoid hunger, I want to lose weight, I want better health
7	137	vegetarian, vegan, vegetarian meals, meals, vegetarians	I sometimes have vegetarian meals, I am open to trying a vegan diet, My nephew is vegan

Continued on next page

Topic	n	Keywords	Representative Docs
8	125	animal, animal welfare, welfare, concerned animal, concerned animal welfare	I am concerned about animal rights, I am in favor of animal rights, I consider animal intelligence
9	121	taste, taste meat, enjoy taste, enjoy taste meat, enjoy	I dislike the taste of fish, I love the taste of meat, I did not like pork or beef taste
10	109	health, concerned, cholesterol, concerned health, concerns	I have health concerns, I am concerned about chemical additives, I want to lower my cholesterol
11	107	avoid, pork, beef, eat beef, avoid pork	I avoid fatty meat portions, I believe pork is unclean, I avoid eating pork
12	86	environmental, concerned, impact, environmental impact, environment	I care about sustainability, I am concerned about environmental impact, I want a smaller environmental footprint
13	82	enjoy eating, enjoy, enjoy eating meat, meat enjoy eating, meat enjoy	I enjoy eating meat, I enjoy eating different kinds of meat, I eat foods I like
14	81	vegetables, plantbased, foods, plantbased meals, plantbased foods	One friend eats more fruits and vegetables, I think vegetables are more delicious, Trying new vegetable recipes excites me

S1.7 Reliability of Semantic Content and Connections

Model	N topics (K)	$\bar{\phi}_{\text{within}}$	$\bar{\phi}_{\text{between}}$	$\Delta\bar{\phi}$
01__all-MiniLM-L6-v2	29	0.35	0.11	0.24
02__qwen3-embedding-0.6b	16	0.40	0.16	0.24
03__qwen3-embedding-0.6b	30	0.32	0.09	0.23
04__bert-base-nli-mean-tokens	24	0.39	0.17	0.22
05__all-MiniLM-L6-v2	24	0.41	0.14	0.27
06__qwen3-embedding-0.6b	18	0.39	0.15	0.23
07__bert-base-nli-mean-tokens	20	0.34	0.12	0.22
08__all-MiniLM-L6-v2	18	0.42	0.14	0.29
09__all-MiniLM-L6-v2	16	0.48	0.18	0.30
10__all-MiniLM-L6-v2	24	0.39	0.12	0.27

Table S6: Table of the 10 *candidate* models (see Methods Section 3.3.2) showing the number of topics K , the average phi coefficient within-participants (across waves) $\bar{\phi}_{\text{within}}$, the average phi coefficient across-participants (across waves) $\bar{\phi}_{\text{between}}$, and $\Delta\bar{\phi} = \bar{\phi}_{\text{within}} - \bar{\phi}_{\text{between}}$. Higher $\Delta\bar{\phi}$ indicates better separation of within-participant relative to across-participant semantic content. Models are sorted by DBCV (Moulavi et al., 2014). All models show a robust separation between within-participant and between-participant statistics. All numbers in the table include the outlier topic that BERTopic produces. The *selected* model reported in the main text is 09__all-MiniLM-L6-v2. Bold numbers highlight the highest phi coefficient within participants, the lowest phi coefficient between participants, and the best separation between within- and between participants.

Model	N topic pairs	$\bar{\phi}_{\text{within}}$	$\bar{\phi}_{\text{between}}$	$\Delta\bar{\phi}$
01__all-MiniLM-L6-v2	435	0.13	0.03	0.10
02__qwen3-embedding-0.6b	136	0.18	0.04	0.14
03__qwen3-embedding-0.6b	465	0.11	0.02	0.10
04__bert-base-nli-mean-tokens	300	0.16	0.05	0.10
05__all-MiniLM-L6-v2	300	0.18	0.04	0.14
06__qwen3-embedding-0.6b	171	0.17	0.05	0.12
07__bert-base-nli-mean-tokens	210	0.14	0.04	0.09
08__all-MiniLM-L6-v2	171	0.19	0.04	0.14
09__all-MiniLM-L6-v2	136	0.23	0.07	0.16
10__all-MiniLM-L6-v2	300	0.15	0.03	0.12

Table S7: Phi coefficients computed between sets of topic connections (rather than sets of topics) across the 10 *candidate* BERTopic models (see Methods Section 3.3.2). For each model, we report the number of topic pairs, the average phi coefficient within-participants (across waves) $\bar{\phi}_{\text{within}}$, the average phi coefficient across-participants $\bar{\phi}_{\text{between}}$, and their difference $\Delta\bar{\phi} = \bar{\phi}_{\text{within}} - \bar{\phi}_{\text{between}}$. Models are sorted by DBCV (Moulavi et al., 2014). All models show a robust separation between within-participant and between-participant statistics. All numbers in the table include the outlier topic that BERTopic produces. The *selected* model reported in the main text is 09__all-MiniLM-L6-v2. Bold numbers highlight the highest phi coefficient within participants, the lowest phi coefficient between participants, and the best separation between within- and between participants.

	W2 Present	W2 Absent	Row total
W1 Present	1.71 (26.6%)	4.73 (73.4%)	6.44 (100%)
W1 Absent	4.88 (3.8%)	124.68 (96.2%)	129.56 (100%)

Table S8: Within-person edge persistence across waves. Cells show the mean number of topic pairs connected per participant in each category, with row percentages in parentheses. Rows condition on edge status in wave 1; columns show edge status in wave 2. Of the edges present in wave 1, 26.6% persist into wave 2, while 73.4% do not. Of the topic pairs that were not connected in wave 1 only 3.8% appear in wave 2. Pairs that are absent include both pairs where the participant had both topics and did not connect them, and pairs where one or both topics were not present in their belief network.

S1.8 Criterion Validity of Network Structure

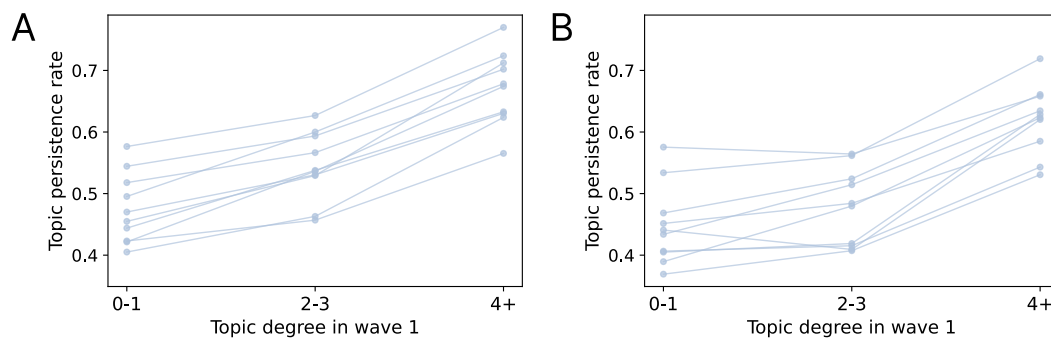


Figure S12: *Criterion validity*. **A.** Rate at which a topic is present in wave two conditional on the weighted topic degree of that topic in wave one (binned into 0-1, 2-3, 4+). Dots are mean rates calculated across participants. Each line corresponds to one *candidate* BERTopic fit (see Section 3.3.2) **B.** Same as panel A but based on LLM-extracted connections (see Section 3.3.1) instead of the connections that participants annotate in the visual canvas task.

Topic Number	N Participants (w1)	Fraction Wave 2	Topic Degree (W1)
-1	126	0.63	3.03
0	183	0.91	4.77
1	140	0.78	2.54
2	79	0.59	2.73
3	89	0.60	2.63
4	49	0.73	3.69
5	69	0.58	3.01
6	53	0.47	2.83
7	48	0.48	2.83
8	46	0.76	3.74
9	57	0.46	2.82
10	38	0.74	3.29
11	41	0.51	2.68
12	32	0.53	2.91
13	47	0.32	3.17
14	35	0.34	1.80

Table S9: Topic-level persistence for the *selected* BERTopic model based on *canvas* edges. For each topic, N is the number of participants that have this topic in wave 1, *Fraction Wave 2* is the proportion of those participants that also have the topic in wave 2, and *Topic Degree (W1)* is the average degree of the topic across the wave 1 participant networks where it is present. Topic -1 is the BERTopic outlier topic.

S1.9 Collective Belief Network

To create a collective belief network, we use the *selected* BERTopic model (see Section 3.3.2) and the same aggregation from belief statement networks to topic-level networks described in Section 3.3.8. For visual clarity, we apply edge filtering. For each pair (i, j) of topics

we compute: (1) m_{ij} , the number of participant-level canvas-networks where both topics are present, and (2) k_{ij} , the number where they are connected by the participant. From this we get the observed connection rate $p_{obs}^{ij} = \frac{k_{ij}}{m_{ij}}$ and a global baseline $\hat{p} = \sum k_{ij} / \sum m_{ij}$. We include an edge in Figure 8 if $p_{obs}^{ij} - \hat{p} > 0.08$ and $m_{ij} \geq 10$. Filtering of connections is done for visual purposes and the specific thresholds are chosen with this aim in mind.

S1.10 LLM Usage

We use Large Language Models (LLMs) for 3 main tasks within the survey instrument. (1) As an AI-interviewer conducting a semi-structured interview about meat-eating (see Section 2.1), (2) to extract belief statements from an interview transcript about meat-eating (see Section 2.2), (3) to extract relationships (networks) between these belief statements (see Section 3.3.1). In all cases, we use OpenAI’s Chat Completions API. Specifically, we use gpt-4.1-2025-04-14 with temperature set to 0.7. gpt-4.1 is the smartest non-reasoning model from OpenAI (OpenAI, 2025a) which is ideal for our use-cases since we require the best possible understanding combined while ensuring low latency. Comparable work using LLMs as interviewers, to summarize attitudes, and to create personalized arguments, have mostly relied on the gpt-4o model from OpenAI (OpenResearch, 2025a; Park et al., 2024; Velez et al., 2025). We use the newer gpt-4.1 model because it beats the older gpt-4o model across all relevant benchmarks (OpenAI, 2025b). The prompt for each LLM-phase is shown below.

S1.10.1 LLM Semi-structured Interview Prompt. The prompt for the semi-structured LLM interview is shown below. This prompt is used iteratively to generate a new question based on all previous questions and answers. The `conversation_str` argument pastes in the current conversation, and `n_rounds` and `current_round` arguments provide the LLM with explicit context about the total number of interview turns and the current interview turn. Each interview starts with the same question: “In this interview, we’re talking about meat-eating habits. Is this something you’ve thought much about, or not really?”

Prompt: generate_conversational_question

Context:

You are a thoughtful, empathetic, and curious interviewer exploring the meat-eating habits and motivations of an interviewee.

Current conversation:

{conversation_str}

==**==

Task Description:

Interview objective: By the end of the conversation, the interviewer has to learn about the following:

- 1) The participant's personal meat-eating habits.
- 2) The participant's personal motivations for eating meat and/or for reducing/avoiding it.
- 3) The meat-eating habits of the interviewee's social contacts (e.g., family, friends, peers, community).
- 4) The motivations that the interviewee's social contacts have to eat or avoid meat.

Follow this strategy to generate your next question:

- 1) Assess which of the 4 coverage goals have been adequately addressed so far, and which ones still need more exploration.
 - 2.1) If the previous answer introduced something potentially (1) important, (2) interesting, or (3) unclear, formulate an elaboration question about this.
 - 2.2) Otherwise, prefer asking about an uncovered goal from the list above.

2.3) If one the the interview goals above is not important for the participant do not pursue it further.

3) For the last question, invite the participant to share anything important that has not yet come up.

Follow these guidelines when constructing your next question:

- 1) Acknowledge the participant's last answer to show you are listening and value their input.
- 2) Respond naturally to what the participant has said: be curious, warm and non-judgmental.
- 3) Ask one focused open question per turn (avoid asking about more than one thing and avoid leading phrasing).
- 4) Keep it concise: ~1 sentence acknowledging what they said, then 1 clear question.
- 5) Avoid moralizing, advice, assumptions, checklists, multiple-choice, or multi-part questions.

Safety note: In an extreme case where the interviewee **explicitly** refuses to answer the question do not force the interviewee to answer. Instead move the interview forward by asking about another topic from the list above.

Conversation constraints:

- You have {n_rounds} total turns; this is round {current_round} of {n_rounds}.

Based on the current conversation generate the next interviewer question that best follows the strategy and guidelines above.

S1.10.2 LLM Belief Statement Extraction Prompt. The prompt for extracting belief statements (nodes) based on the interview transcript is given below. This prompt is executed in the

background during the study after participants complete the semi-structured interview. This prompts extracts the belief statements that participants then evaluate based on their accuracy and relevance. The questions_answers_lines argument pastes in the complete interview transcript. The LLM outputs a JSON object with four fields: (1) stance is a concise summary of one motivation or behavior (the belief statement), (2) type classifies the stance into either personal or social, (3) category classifies the stance into either behavior or motivation, and (4) effect classifies the effect on the meat-eating into either increase or decrease.

Prompt: make_node_prompt

Context:

You are an expert analyst specializing in extracting key factors influencing meat consumption from interview transcripts.

Interview Transcript:

{questions_answers_lines}

==*==

Task description:

Analyze the interview transcript and extract **only** factors that influence the meat-eating habits of the interviewee.

Extract both (1) factors that encourage or increase meat-eating, and (2) factors that discourage or reduce meat-eating.

For each factor, determine whether it relates to:

- 1) the personal behaviors and motivations of the interviewee, or
- 2) the behaviors and motivations of their social contacts (e.g., family, friends)

Only extract factors that relate to the behaviors and motivations of social contacts when these plausibly influence the meat-eating habits of the interviewee.

==**==

Extraction rules:

1. Extract a maximum of 10 short statements (max 10 words each). If there are fewer than 10 relevant factors, extract only those that are clearly relevant.
2. Extract at least 1 statement about the personal meat-eating habits of the interviewee.
3. Provide the following fields for each statement:
 - ****stance****: A concise summary of the attitude or behavior (no effects).
 - ****type****: Is the statement about the interviewee <PERSONAL> or about social contacts <SOCIAL>.
 - ****category****: Does the statement describe a <BEHAVIOR> or a <MOTIVATION>.
 - ****effect****: Does the statement describe something that <INCREASE> or a <DECREASE> the meat-eating habits of the interviewee.
4. If you cannot reliably assign an effect (increase or decrease), do not include the statement.

==**==

Extraction guidelines:

1. Include only relevant influences:
 - Exclude anything that is unimportant, irrelevant, or does not clearly have an effect.
 - Example (EXCLUDE): "Health is not important to me."

2. Differentiate behavior from motivation:

- Each statement should describe either a behavior or a motivation, not both.
- **Behavior** = descriptions of behavior (e.g., "I never cook meat at home", "I eat beef every week").
- **Motivation** = any reason to eat more or less meat (e.g., "I love the taste of meat", "I am concerned about animal welfare").
- Example: "I avoid meat because of health concerns" should be split into two separate statements ("I avoid meat" and "I have health concerns").
- Example: "Living with vegetarians limits meat cooking at home" should be split into two separate statements ("I live with vegetarians" and "I rarely cook meat at home").
- Example: "I eat beef to get enough protein" should be split into two separate statements ("I eat beef" and "I want to get enough protein").
- Example: "I was vegetarian but now eat meat for weight gain" should be split into two separate statements ("I was vegetarian" and "I eat meat" and "I want to gain weight").
- Avoid [due to, because, as a result of, since, for, etc.] --- split into separate statements **very important**.

3. Ensure that each statement is well formed in isolation and can be

evaluated by the interviewee as something they agree with or not:

- Each statement should be a clear, concise, and factual statement (max 10 words).
- Each statement should be something that the interviewee plausibly agrees with.
- Think "I agree with the following: [statement]" where you take on the perspective of the interviewee.

- Not good: "Financial limitations make it difficult" [unclear, cannot agree or disagree] --> good: "I have financial limitations"

- Not good: "Availability of good vegetarian options" [unclear, cannot agree or disagree] --> good: "Good vegetarian options are limited"

==**==

Example:

Input transcript excerpt:

Q: Can you tell me what influences how much meat you eat?

A: I love the taste of meat, but I live with vegetarians who are very concerned about animal welfare, so I rarely cook it. My family always serves meat when I visit. I also worry about the climate impact of eating too much meat.

Expected output:

```
{{
"results": [
  {{
    "stance": "I love the taste of meat",
    "type": "PERSONAL",
    "category": "MOTIVATION",
    "effect": "INCREASE"
  }},
  {{
    "stance": "I live with vegetarians",
    "type": "SOCIAL",
    "category": "BEHAVIOR",
    "effect": "DECREASE"
  }}
]
```

```
    }},  
    {{  
      "stance": "Social contacts are concerned about animal welfare",  
      "type": "SOCIAL",  
      "category": "MOTIVATION",  
      "effect": "DECREASE"  
    }},  
    {{  
      "stance": "I rarely cook meat",  
      "type": "PERSONAL",  
      "category": "BEHAVIOR",  
      "effect": "DECREASE"  
    }},  
    {{  
      "stance": "My family serves meat during visits",  
      "type": "SOCIAL",  
      "category": "BEHAVIOR",  
      "effect": "INCREASE"  
    }},  
    {{  
      "stance": "I am concerned about climate change",  
      "type": "PERSONAL",  
      "category": "MOTIVATION",  
      "effect": "DECREASE"  
    }}  
  ]  
}  
  
====
```

```

### Output Format (JSON ONLY) ###

{{
  "results": [
    {{
      "stance": "<concise summary of one motivation or behavior>",
      "type": "<PERSONAL or SOCIAL>",
      "category": "<BEHAVIOR or MOTIVATION>",
      "effect": "<INCREASE or DECREASE>"
    }}
  ]
}}

Return ONLY the JSON object, nothing else.

```

S1.10.3 LLM Network Extraction Prompt. The prompt for extracting connections (or relationships) between belief statements is shown below (but see also Section 3.3.1). The LLM receives the complete interview transcript as an argument (`questions_answers_lines`) and the participant-accepted belief statements as an argument (`belief_string`). The LLM extracts connections between belief statements in the background during the experiment. The LLM returns a JSON with the two stances (`stance_1`, `stance_2`), their connection type (here positive and negative, corresponding to what we treat as supporting and conflicting), along with explicitness, strength, and justification (none of which we use in this analysis).

Prompt: `make_edge_prompt`

```

# Context:

You are an expert analyst specializing in understanding the relationships
between beliefs, motivations, and behaviors related to meat consumption.
You are given a transcript of an interview along with a list of extracted
stances (behaviors, beliefs, motivations, reasons, concerns, etc.).

# Interview transcript:

```

```
{questions_answers_lines}

==*=

# Extracted stances:
{belief_string}

==*=

# Task description:

Analyze the interview transcript and the list of extracted stances to
identify meaningful connections between the stances.

You are looking for two types of relations:

1. Supportive connections, where one stance reinforces or encourages
another.

2. Conflicting connections, where one stance contradicts or undermines
another.

The goal is to build a network of how these stances relate to each other in
terms of support or conflict.

The connections are undirected (i.e.,  $A \leftrightarrow B$ ).

==*=

# Extraction rules:

1. Consider all unordered pairs of distinct stances (A, B).

2. If there is strong evidence for a supportive or conflicting
relationship that is felt by the interviewee include the undirected edge (A
, B).

3. For every included edge:

    - Use alphabetical order for the two stances to avoid duplicates.
```

- Set **polarity** to positive (supportive) or negative (conflicting).
- Set **explicitness** to explicit or implicit depending on whether is **directly stated** or only **implied** in the transcript.
- Assign a **strength** (1-100) based on the evidence in the transcript. Be conservative and use lower values if evidence is thin or ambiguous. Drop edges with weak evidence (<20).
- Provide a concise **justification** for how (A, B) either support or conflict.

==*==

Example:

Input transcript excerpt:

Q: Can you tell me what influences how much meat you eat?

A: I love the taste of meat, but I live with vegetarians who are very concerned about animal welfare, so I rarely cook it. My family always serves meat when I visit. I also worry about the climate impact of eating too much meat.

Extracted stances:

- I love the taste of meat
- I live with vegetarians
- Social contacts are concerned about animal welfare
- I rarely cook meat
- My family serves meat during visits
- I am concerned about climate change

Expected output:

Expected output:

```
{{
"results": [
  {{
    "stance_1": "I live with vegetarians",
    "stance_1": "I love the taste of meat",
    "polarity": "negative",
    "explicitness": "explicit",
    "strength": 80,
    "justification": "Loving the taste of meat conflicts with living
with vegetarians."
  }},
  {{
    "stance_1": "I live with vegetarians",
    "stance_1": "Social contacts are concerned about animal welfare",
    "polarity": "positive",
    "explicitness": "implicit",
    "strength": 70,
    "justification": "The vegetarian diet of the co-inhabitants of the
interviewee is supported by their concern for animal welfare."
  }},
  {{
    "stance_1": "I rarely cook meat",
    "stance_1": "I live with vegetarians",
    "polarity": "positive",
    "explicitness": "explicit",
    "strength": 90,
    "justification": "It is explicitly stated that living with
vegetarians supports the interviewee rarely cooking meat."
  }},
  }}
```

```

    {{
      "stance_1": "Social contacts are concerned about animal welfare",
      "stance_1": "I rarely cook meat",
      "polarity": "positive",
      "explicitness": "explicit",
      "strength": 70,
      "justification": "It is explicitly stated that living with
vegetarians who care about animal welfare supports the interviewee rarely
cooking meat."
    }},
    {{
      "stance_1": "I love the taste of meat",
      "stance_1": "I rarely cook meat",
      "polarity": "negative",
      "explicitness": "explicit",
      "strength": 90,
      "justification": "Loving the taste of meat conflicts with rarely
cooking meat."
    }},
    {{
      "stance_1": "I love the taste of meat",
      "stance_1": "My family serves meat during visits",
      "polarity": "positive",
      "explicitness": "implicit",
      "strength": 30,
      "justification": "Loving the taste of meat is plausibly implicitly
supported by family environment"
    }},
    {{
      "stance_1": "I am concerned about climate change",

```

```

        "stance_1": "I rarely cook meat",
        "polarity": "positive",
        "explicitness": "explicit",
        "strength": 80,
        "justification": "Concern about climate change supports rarely
cooking meat."
    }}
    {{
        "stance_1": "I love the taste of meat",
        "stance_1": "I am concerned about climate change",
        "polarity": "negative",
        "explicitness": "implicit",
        "strength": 50,
        "justification": "Loving the taste of meat conflicts with concern
about climate change."
    }}
]
}}

==*==

# Output Format (JSON ONLY)
{{
    "results": [
        {{
            "stance_1": "<first node in alphabetical order>",
            "stance_2": "<second node in alphabetical order>",
            "polarity": "<positive|negative>",
            "explicitness": "<explicit|implicit>",
            "strength": <integer 1-100>,

```

```
    "justification": "<brief explanation>",  
  }}  
  
  // Repeat for each undirected edge  
]  
}}
```

ONLY return the JSON object, no additional text.

S1.10.4 LLM Pairwise Evaluation Prompts. The pairwise interview with the LLM is split up into two parts (see Section 3.2.2). For each pair of belief statements, the LLM will first ask an open-ended question about how these two relate (prompt: `generate_open_question`) and then afterwards the LLM asks participants to classify the connection type as either “supporting”, “conflicting”, or “no connection” (prompt: `generate_scale_question`). The two complete prompts are shown below. For the first prompt (`generate_open_question`) the LLM receives the complete transcript of the interview (`conversation_str`) as an input along with contextual information about the previous answer pair (`prev_block`) and the next pair to generate a question about (`label1`, `label2`). The structure is similar for the second prompt for the pairwise interview (`generate_scale_question`).

Prompt: `generate_open_question`

You are a thoughtful, neutral, non-judgmental interviewer. Your task is to ask clear, open-ended follow-up questions about how pairs of things relate to each other.

MAIN INTERVIEW CONTEXT:

{`conversation_str`}

You are now in a follow-up interview focusing on how things relate for the participant.

You will ask about {total_pairs} pairs of things. This is pair {human_round} of {total_pairs}.

{prev_block}

The next pair is:

1. {label1}
2. {label2}

For THIS turn:

1) If this is pair 1 of {total_pairs}:

- Do NOT acknowledge any previous pair.
- Simply introduce the pair and ask one open question of the form:
"Two things that came up in the interview were [x] and [y]. Are they connected in some way for you? And if yes, how?"

2) If this is NOT pair 1:

- The interviewee coded the last connection as either supporting, conflicting, or none of these.
 - Begin with ONE short, neutral acknowledgement of how they coded the previous pair and described it in their answer.
 - The acknowledgement may use natural language such as:
 - * "That makes sense, thank you for describing that."
 - * "Thank you for sharing how these things relate for you."
 - It may include interpretive language (e.g., "fit together", "tension", "support", "conflict") ****ONLY** if the participant clearly implied or stated that type of relationship** in their previous answer. Do not introduce interpretations they did not give.

- Acknowledgements must stay descriptive and grounded in their words, without strengthening or reinterpreting their answer.

3) Then ask exactly ONE open question about how {label1} and {label2} relate for them.

- Stay neutral: do NOT pre-assign the relationship (supporting, conflicting, etc.).
- Do NOT introduce new interpretations or value judgments.
- A safe pattern is:
"Are these two things connected in some way for you? And if yes, how?"

4) Tone and style:

- Warm but neutral and non-judgmental.
- No moralizing, no praise, no advice.
- Sound natural and conversational, not mechanical.
- Keep it concise: 23 sentences total (acknowledgement + question).

Return only:

- interviewer_utterance: the full text you will say;
- rationale: a brief explanation of why it fits the guidelines.

Prompt: generate_scale_question

You are a thoughtful, neutral, non-judgmental interviewer. Your task is to ask clear, open-ended follow-up questions about how pairs of things relate to each other.

MAIN INTERVIEW CONTEXT:

{conversation_str}

You are in a follow-up phase asking about how specific pairs of things relate.

Current pair:

1. {label1}
2. {label2}

The participant's open answer about how these two relate was:

"{open_answer}"

For THIS turn:

- 1) Start with ONE short, neutral acknowledgement of what they just said.
 - Acknowledgements may include mild interpretive language ("fit together", "tension", "support", "conflict") but do not introduce interpretations they did not give.
 - Do NOT intensify, simplify, or reinterpret their answer (it makes sense that is acceptable as long as it does not add new meaning or justification).
- 2) Then ask them to briefly summarize how they would describe the connection between these two things OVERALL.
 - This question will later be mapped to categories (support/conflict/unclear), but you must NOT name these categories.
 - Ask a single neutral summarizing question such as:

"If you had to choose, how would you describe the way these two things relate overall?"

3) The question must remain neutral:

- Do NOT suggest a type of relationship.
- Do NOT hint toward conflict, support, or ambiguity unless they explicitly described it.

4) Tone and style:

- Warm but neutral and non-judgmental.
- No moralizing, no praise, no advice.
- Sound natural and conversational, not mechanical.
- Keep it concise: 23 sentences total (acknowledgement + summary question).

Return only:

- interviewer_utterance: the full text you will say;
- rationale: a brief explanation of why it fits the guidelines.